

① Faraday's Law of EMI:



An EMF is induced in
a Conductor / Coil when
Magnetic flux linked with it
Changes / being cut due to
Some motion

Magnitude of induced EMF

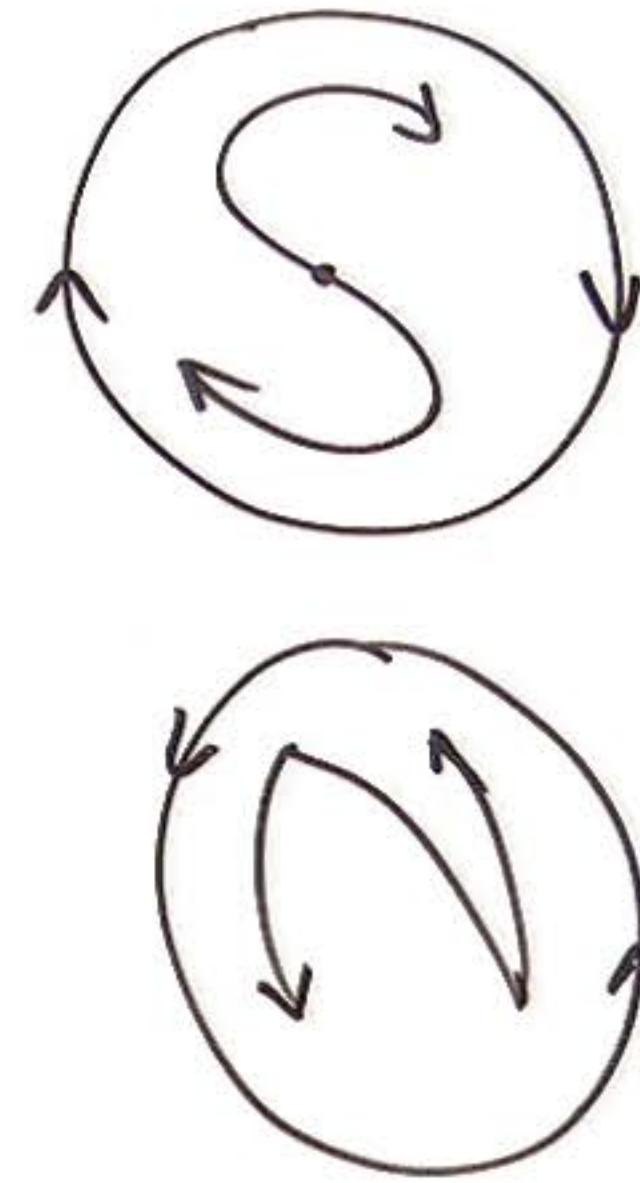
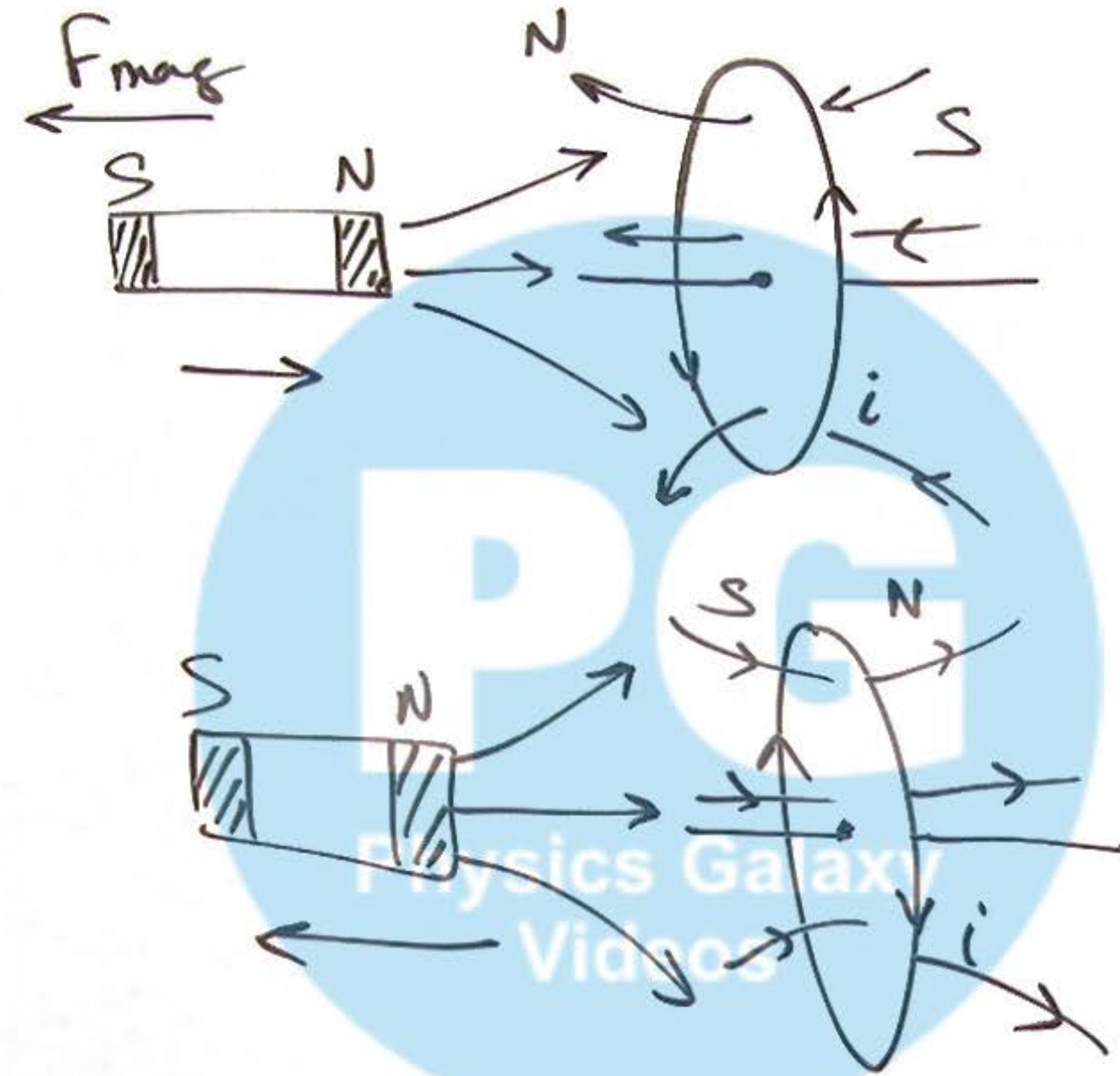
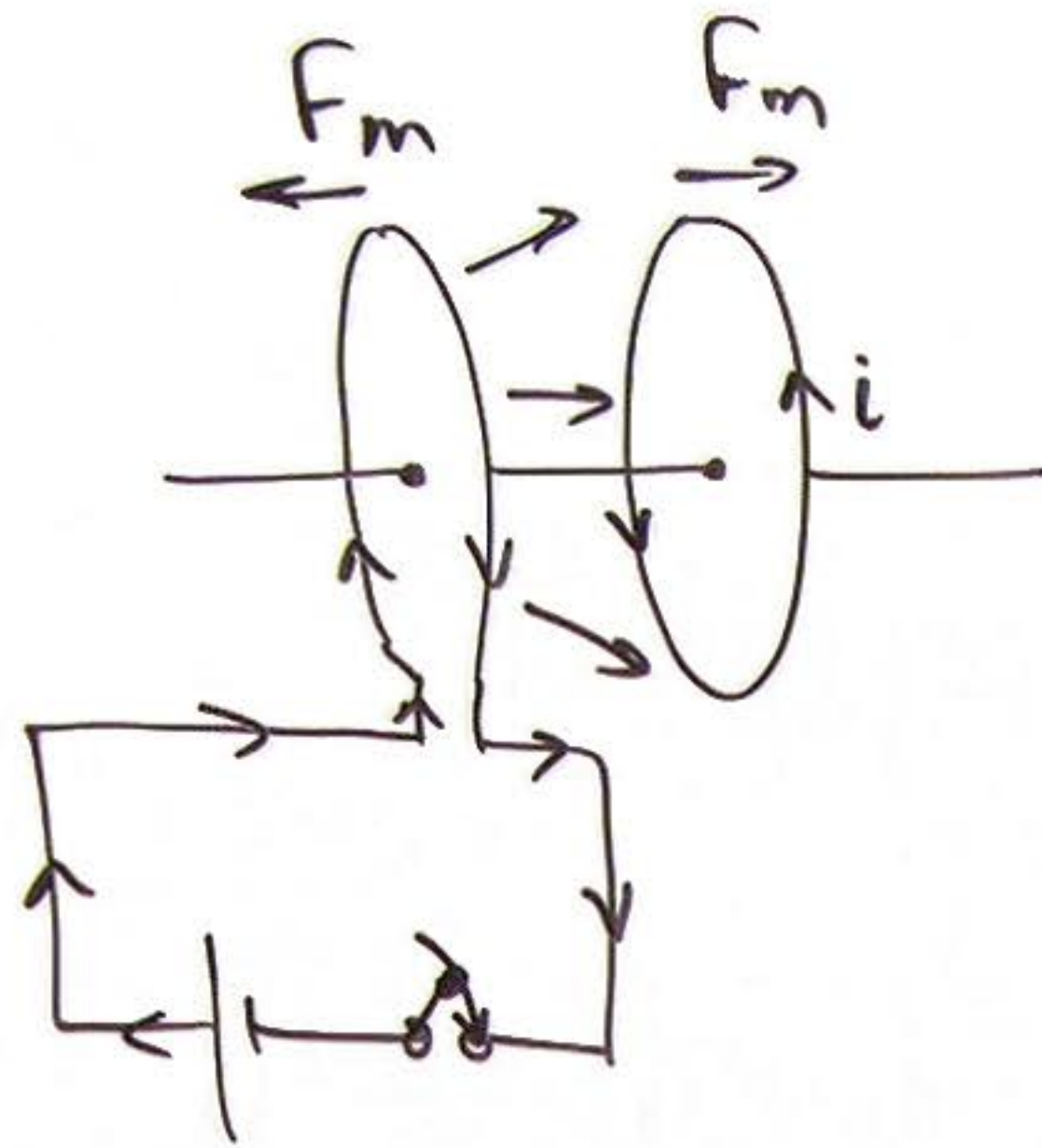
$$e = \left| \frac{d\phi}{dt} \right|$$

$$e = - \frac{d\phi}{dt}$$

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"Lenz's law"

[Always causes of induced EMF are
opposed by the EMF]

② Direction of Induced EMF/Current
in a coil in MF:



② Motional EMF in a moving Conductor:

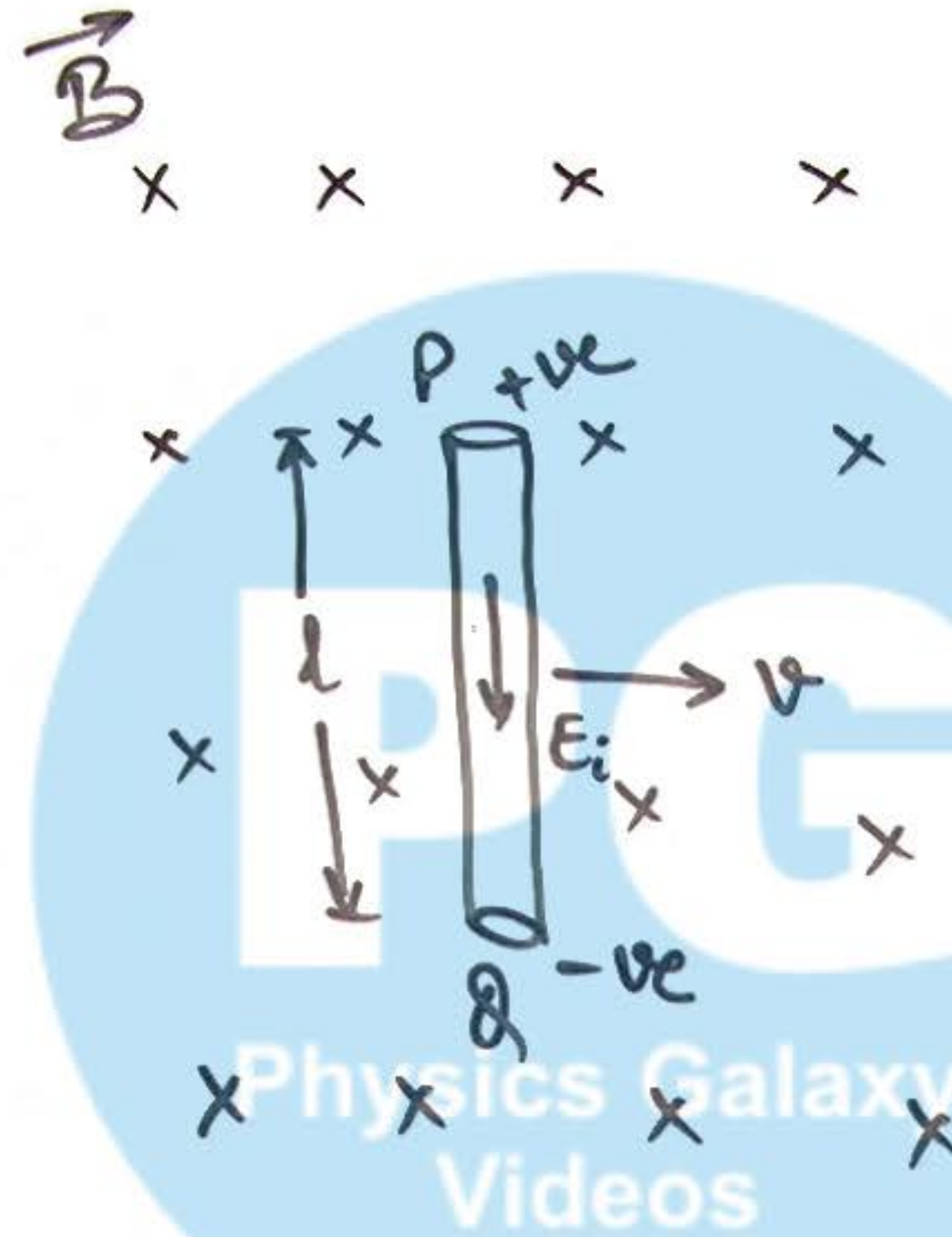
Induced EF

$$E_i = vB$$

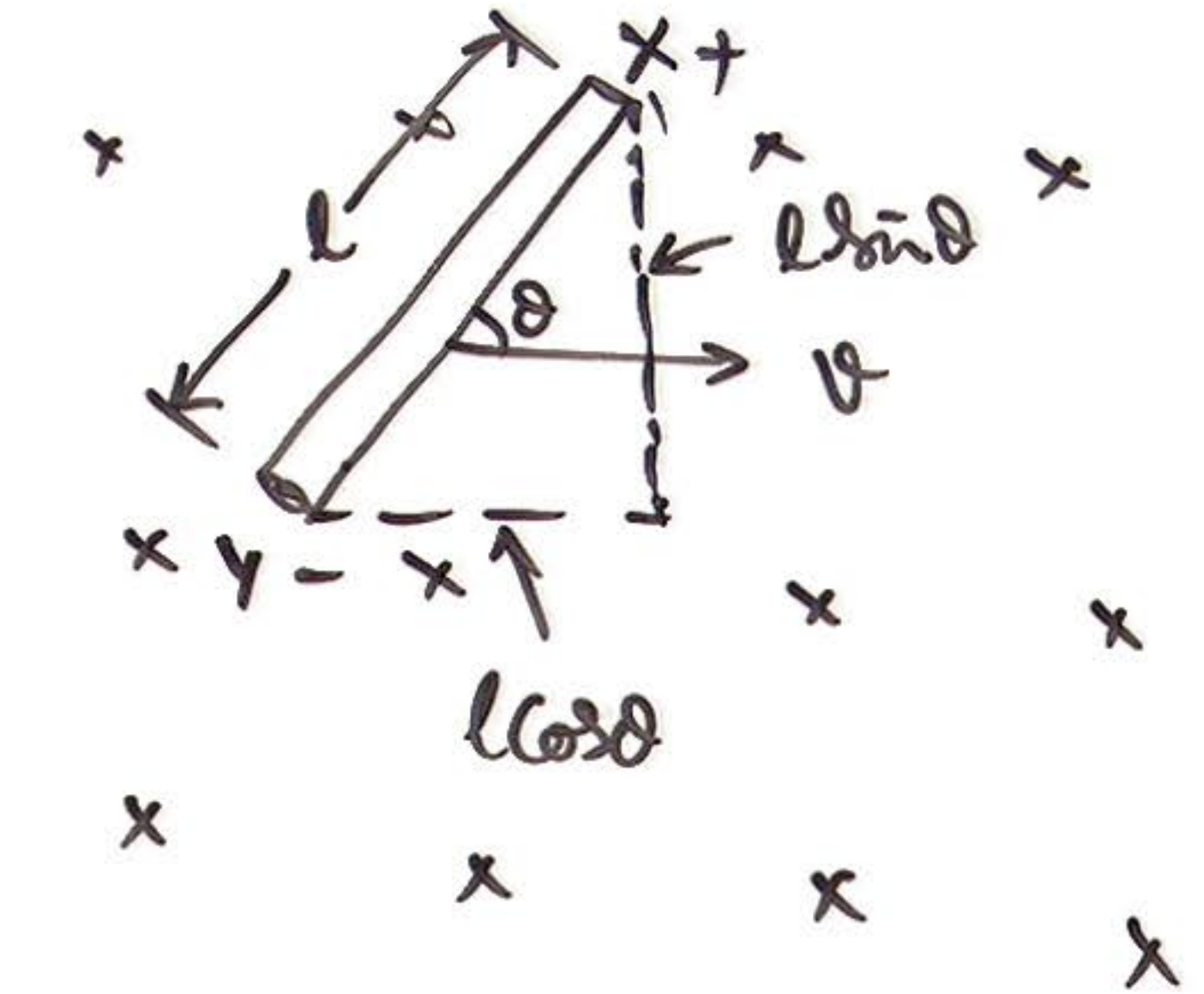
Induced EMF across PQ

$$E_{PQ} = Bvl$$

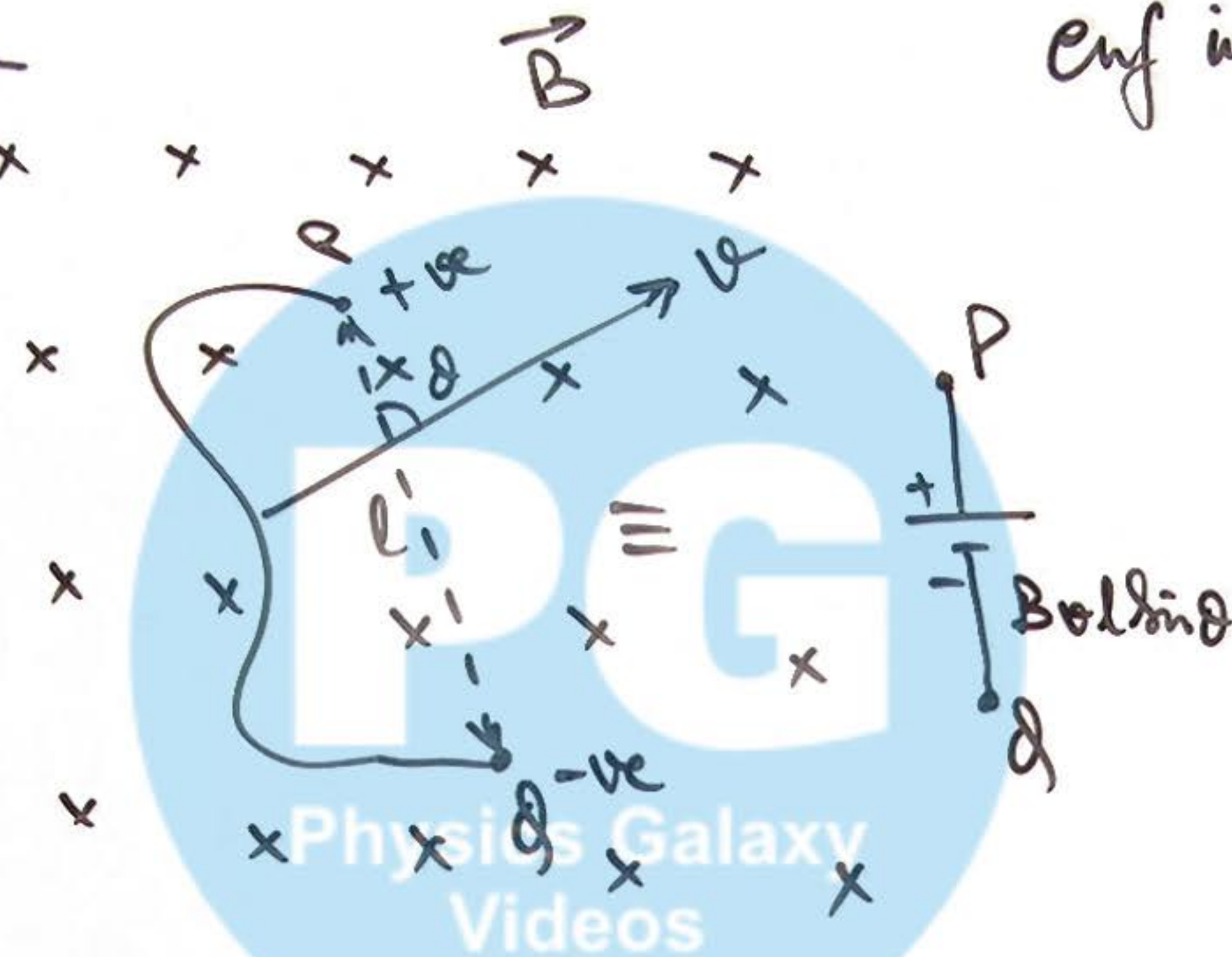
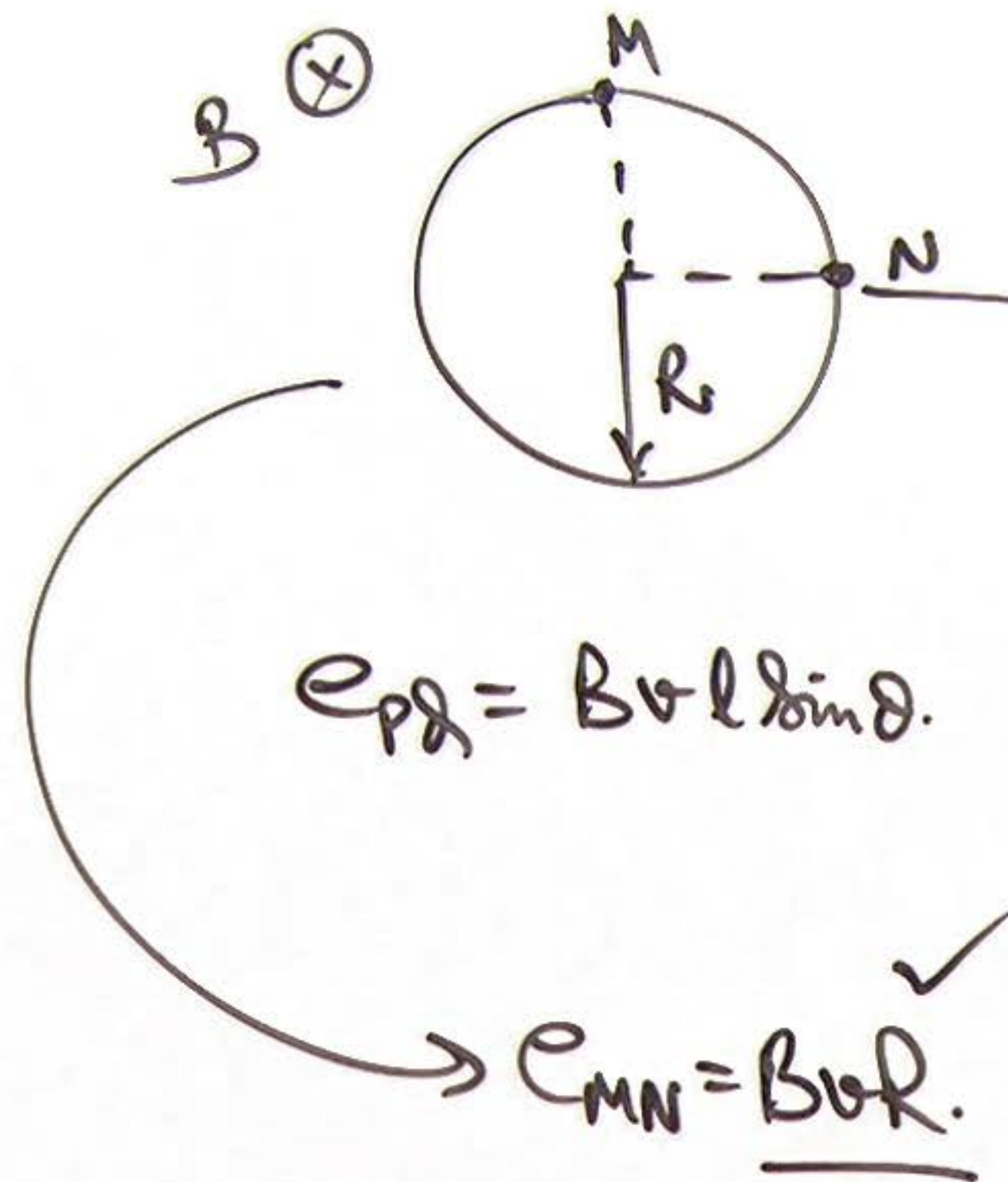
Motional emf



$$E_{xy} = Bvl \sin \theta$$

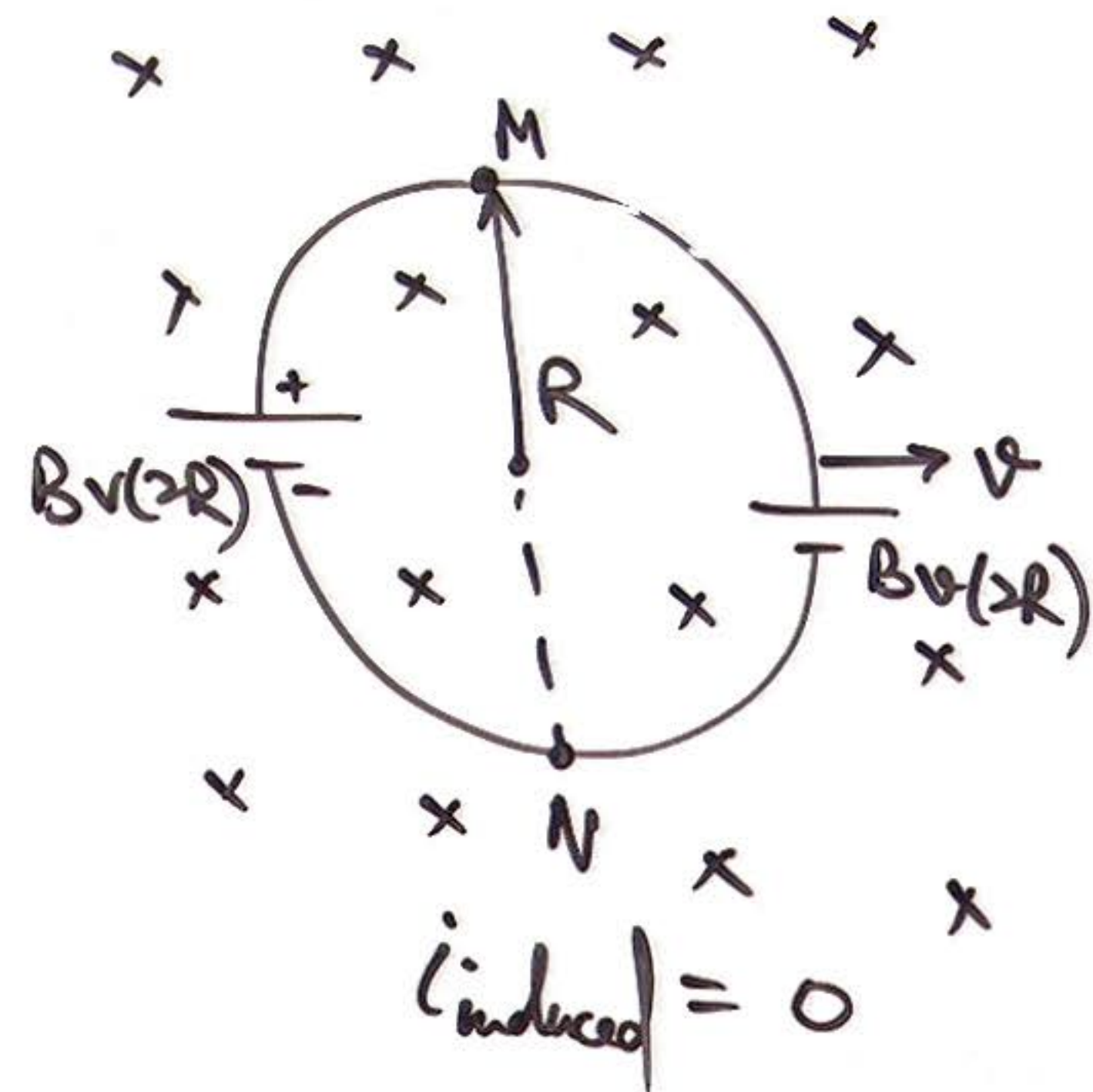


④ Motional EMF in a Random Shaped wire :

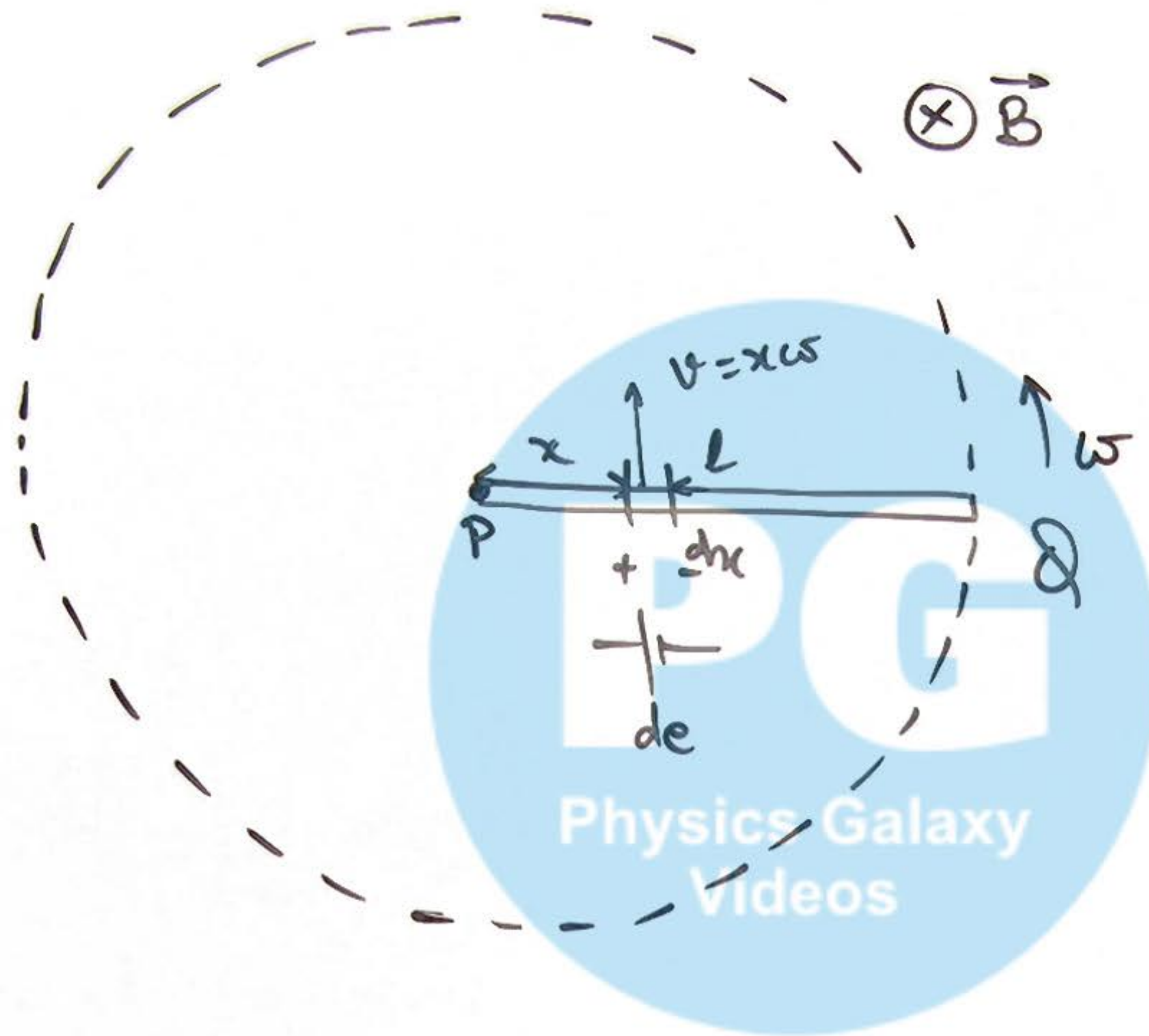


loop emf induced $e_{\text{loop}} = 0$

emf induced across MN $e_{MN} = Bv(2R)$



⑤ Motional EMF in a Rotating Conductor in MF:



Motional emf in element

$$de = B(x\omega) dx$$

$$e_{PQ} = \int de = \int_0^l B\omega x dx = \boxed{\frac{1}{2} B\omega l^2}$$

⑥ Motional EMF in Non-Uniform MF:

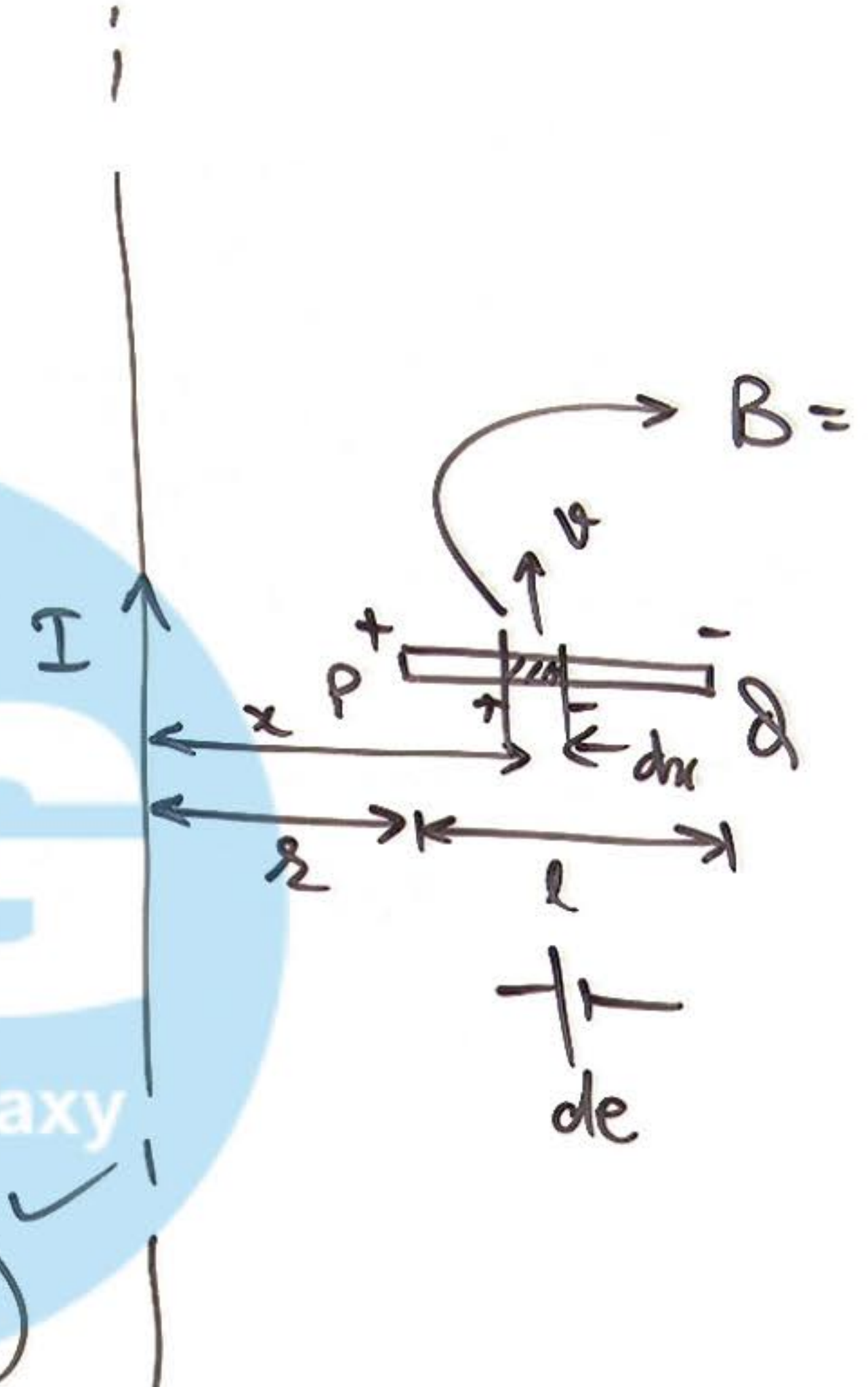
IL:
Find emf induced across PQ.

emf in element

$de = Bv \cdot dx$

$$e_{PQ} = \int de = \int_r^{r+l} \frac{\mu_0 I v}{2\pi x} dx$$

$$= \frac{\mu_0 I v}{2\pi} \ln\left(\frac{r+l}{r}\right)$$



$B = \frac{\mu_0 I}{2\pi x}$

x

P^+

Q^-

dx

r

l

v

I

de

⑦ Motional EMF under External force: $P = F \cdot v_T = \frac{F^2 R}{B^2 l^2} \checkmark \star$

$$i = \frac{Bvl}{R}$$

$$\mathcal{E}_{PQ} = Bvl$$

accⁿ of Conductor PQ

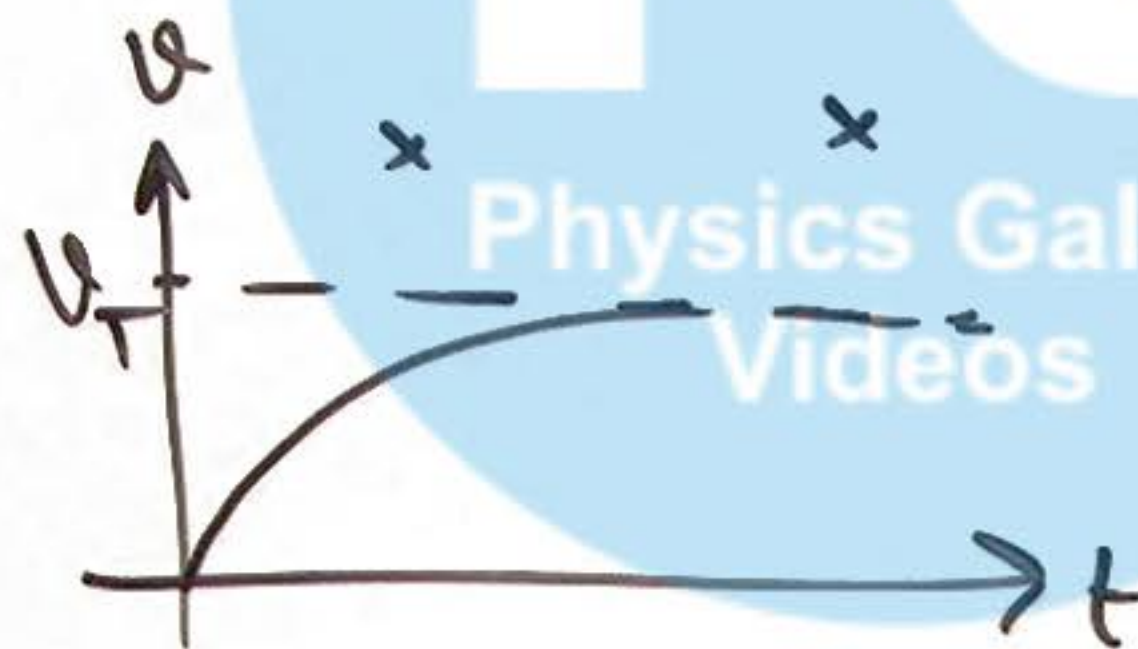
$$a = \frac{F - B il}{m}$$

$$\frac{dv}{dt} \Rightarrow a = \frac{F - \frac{B^2 l^2 v}{R}}{m} = \frac{FR - B^2 l^2 v}{mR}$$

$$F_m = B il$$

$$\Rightarrow F_m = \frac{B^2 l^2 v}{R}$$

$$H = i^2 R = \frac{F^2 R}{B^2 l^2} \checkmark \star$$



$$\Rightarrow v = \frac{FR}{B^2 l^2} \left[1 - e^{-\frac{B^2 l^2 t}{mR}} \right]$$

$$v_T = \frac{FR}{B^2 l^2}$$

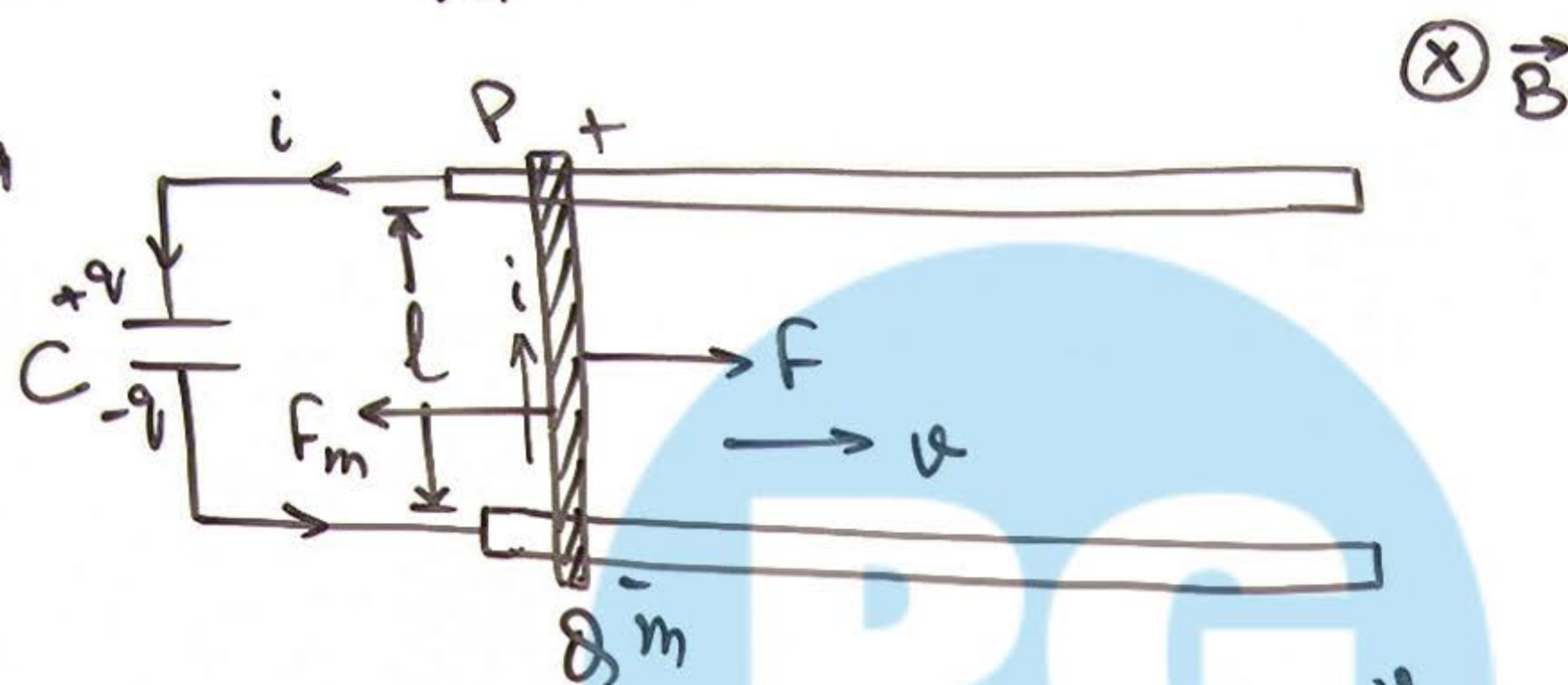
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⑧ Motional EMF with a Capacitor in Circuit

$$q = CV = CB\ell v$$

$$i = \frac{dq}{dt} = CB\ell a$$

$$V_{PQ} = \mathcal{E}_{PQ} = B\ell v$$



$$f_m = B\ell i$$

$$v = at$$

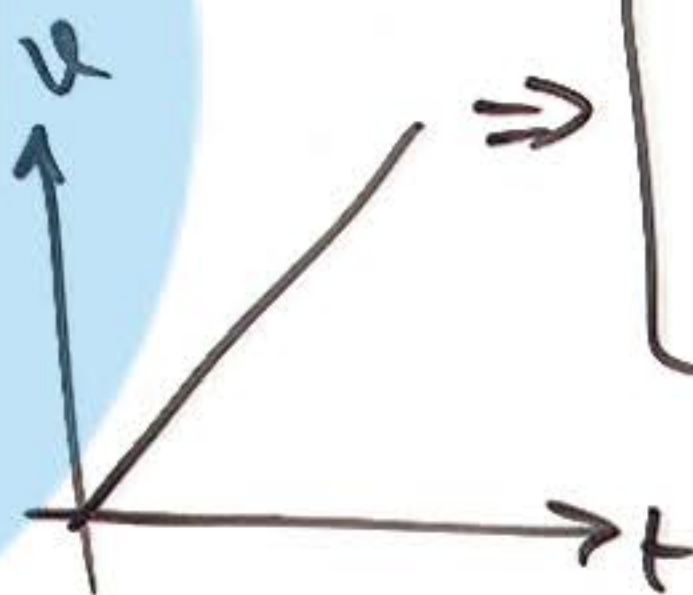
$$v = \sqrt{2as}$$

Accⁿ of Conductor PQ

$$a = \frac{F - B\ell i}{m}$$

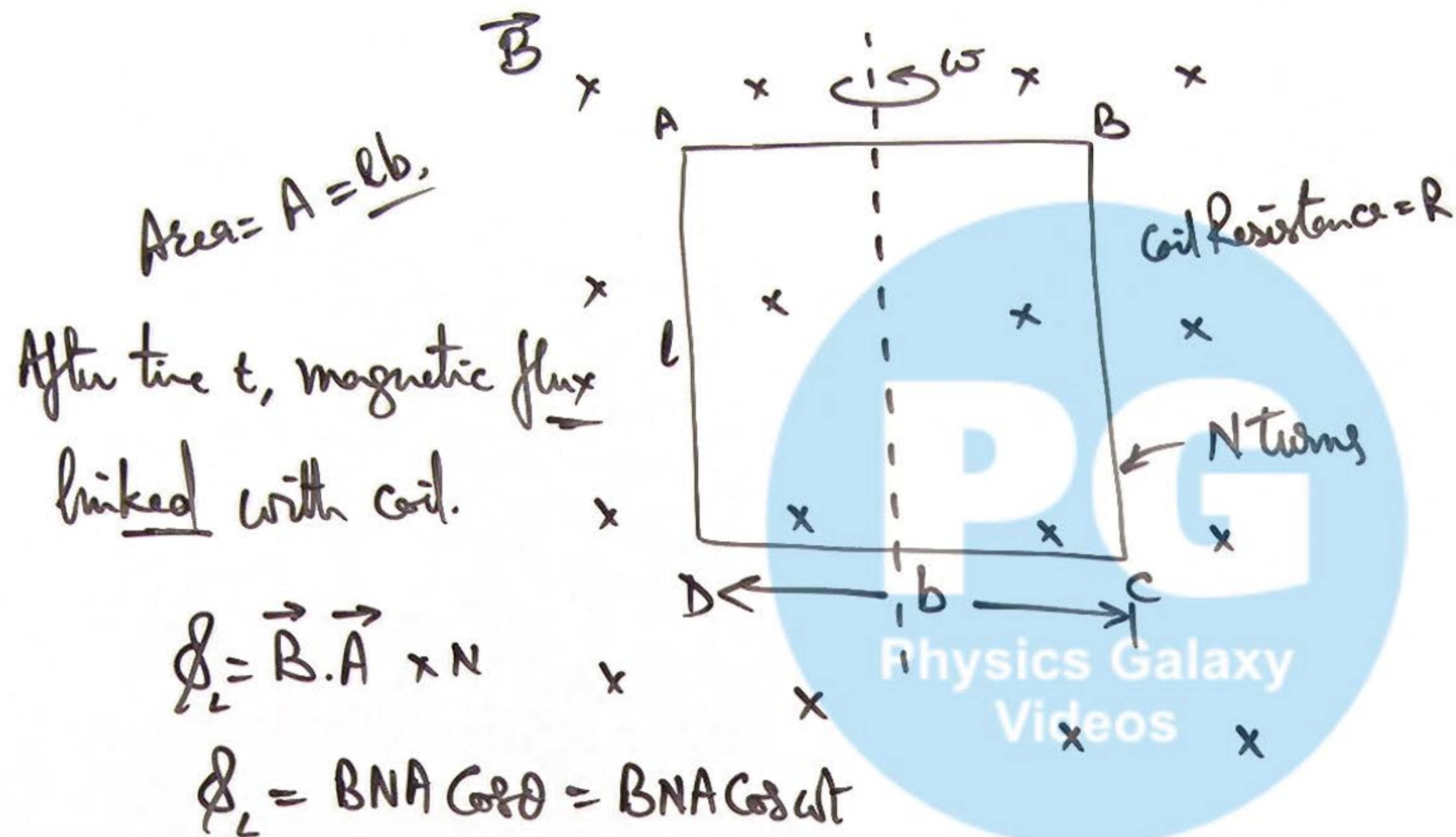
$$ma = F - CB^2\ell^2 a$$

$$\boxed{a = \frac{F}{m + CB^2\ell^2}} \quad [\text{Constant}]$$



⑨ EMF induced in a Rotating Coil in MF:

2dⁿ angle
 $\theta = \omega t$



induced current $i = \frac{e}{R} = \frac{NBA\omega}{R} \sin \omega t$

emf induced in coil

$$e = \left| \frac{d\phi}{dt} \right|$$

$$e = BNA\omega \sin \omega t$$

$$e = e_0 \sin \omega t$$

Alternating emf.

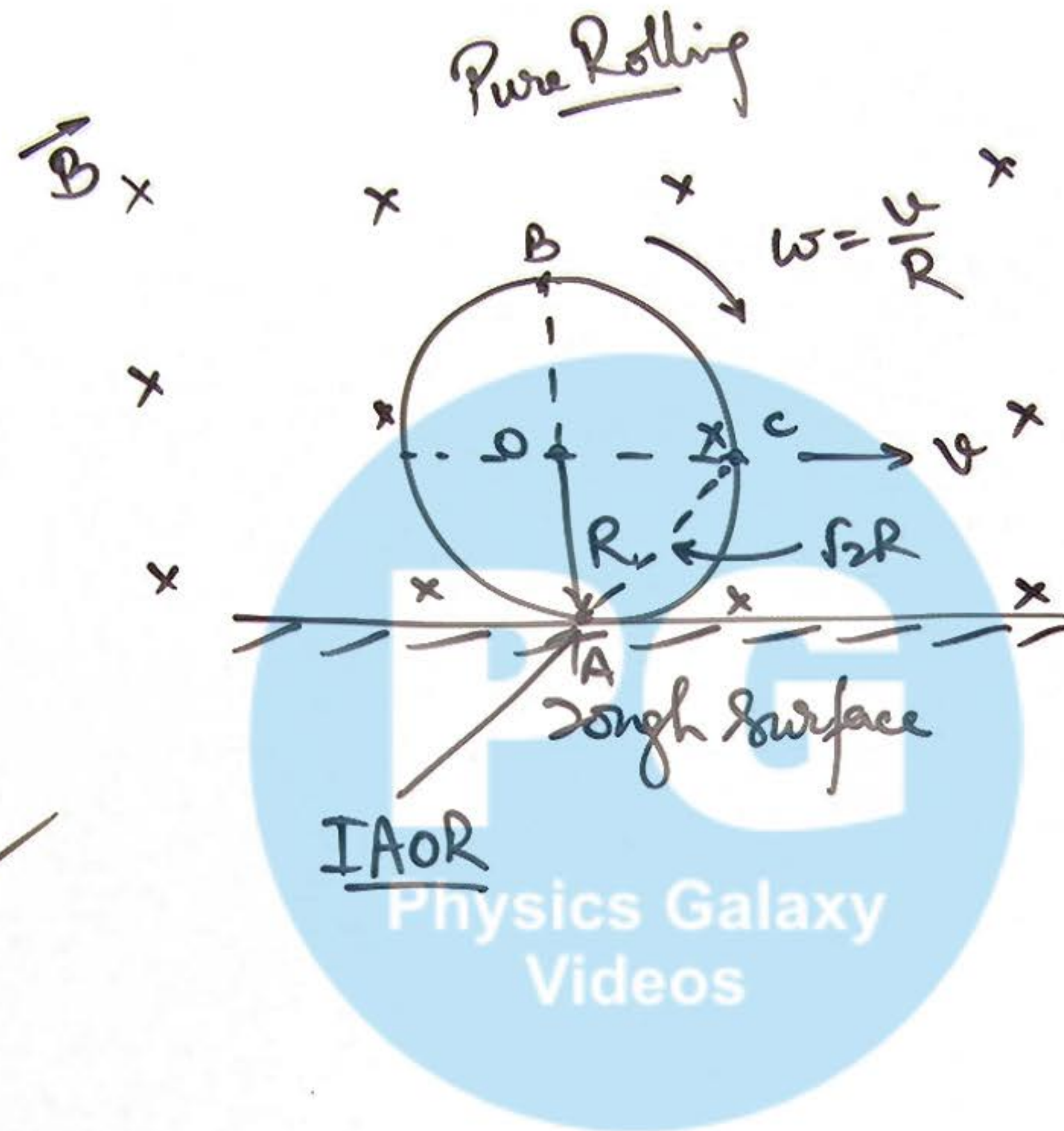
⑩ EMF induced in a Rolling loop :

$$e_{\text{loop}} = 0$$

$$e_{AB} = \frac{1}{2} B \omega l^2$$

$$= \frac{1}{2} B \omega (2R)^2$$

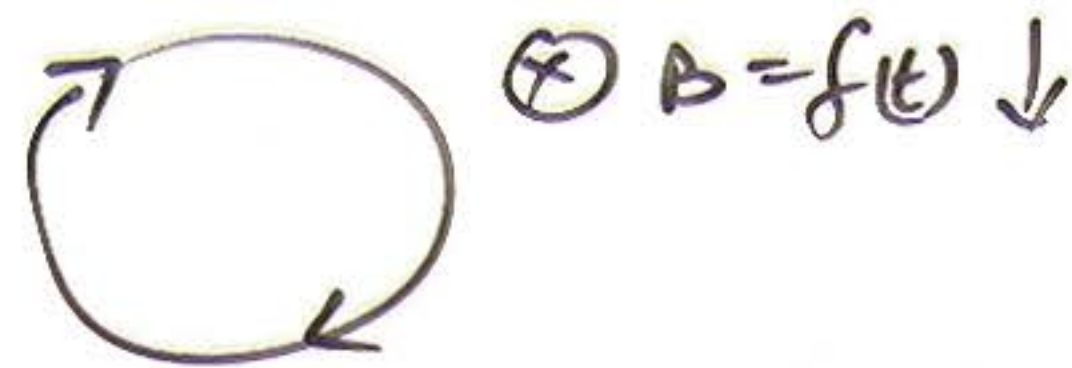
$$= 2 B \omega R^2 \checkmark$$



$$e_{AC} = \frac{1}{2} B \omega (\sqrt{2}R)^2$$

$$e_{AC} = B \omega R^2$$

⑪ EMF induced in Time Varying MF :



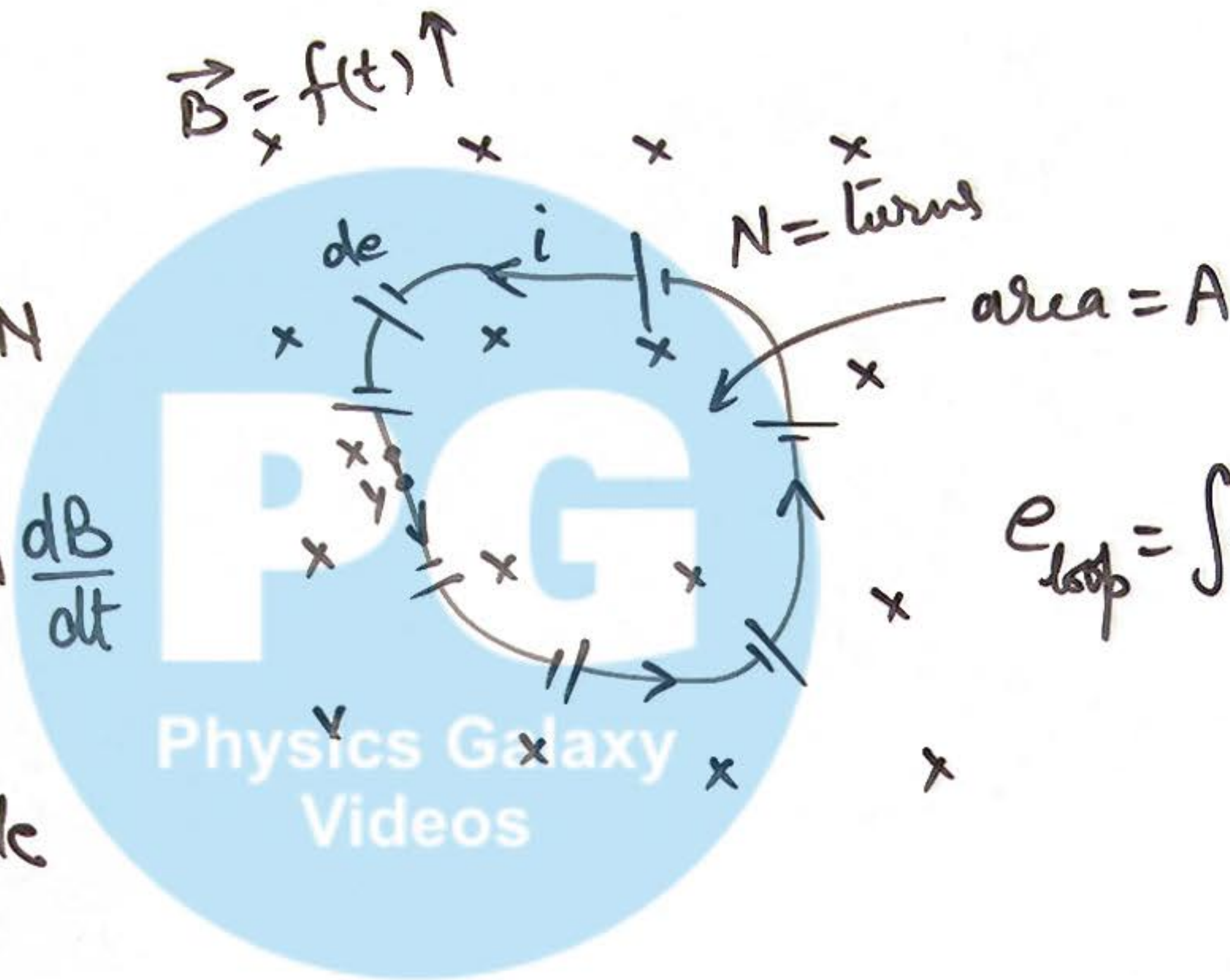
⑫ Direction of induced current in TVMF :
↓
by Lenz's Law.

linked flux through
Coi $\Phi_L = BAN$

$$|V_{xy}| = e = \left| \frac{d\Phi_L}{dt} \right| = NA \frac{dB}{dt}$$

↑
loop emf

↘
 $\int de$

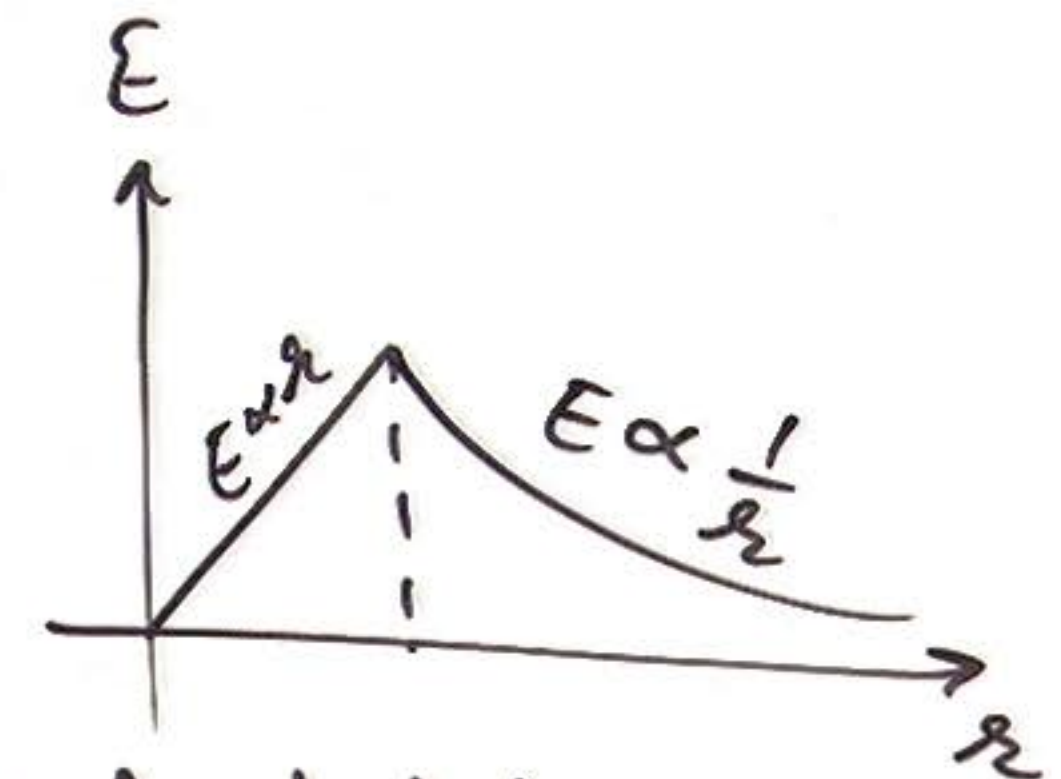
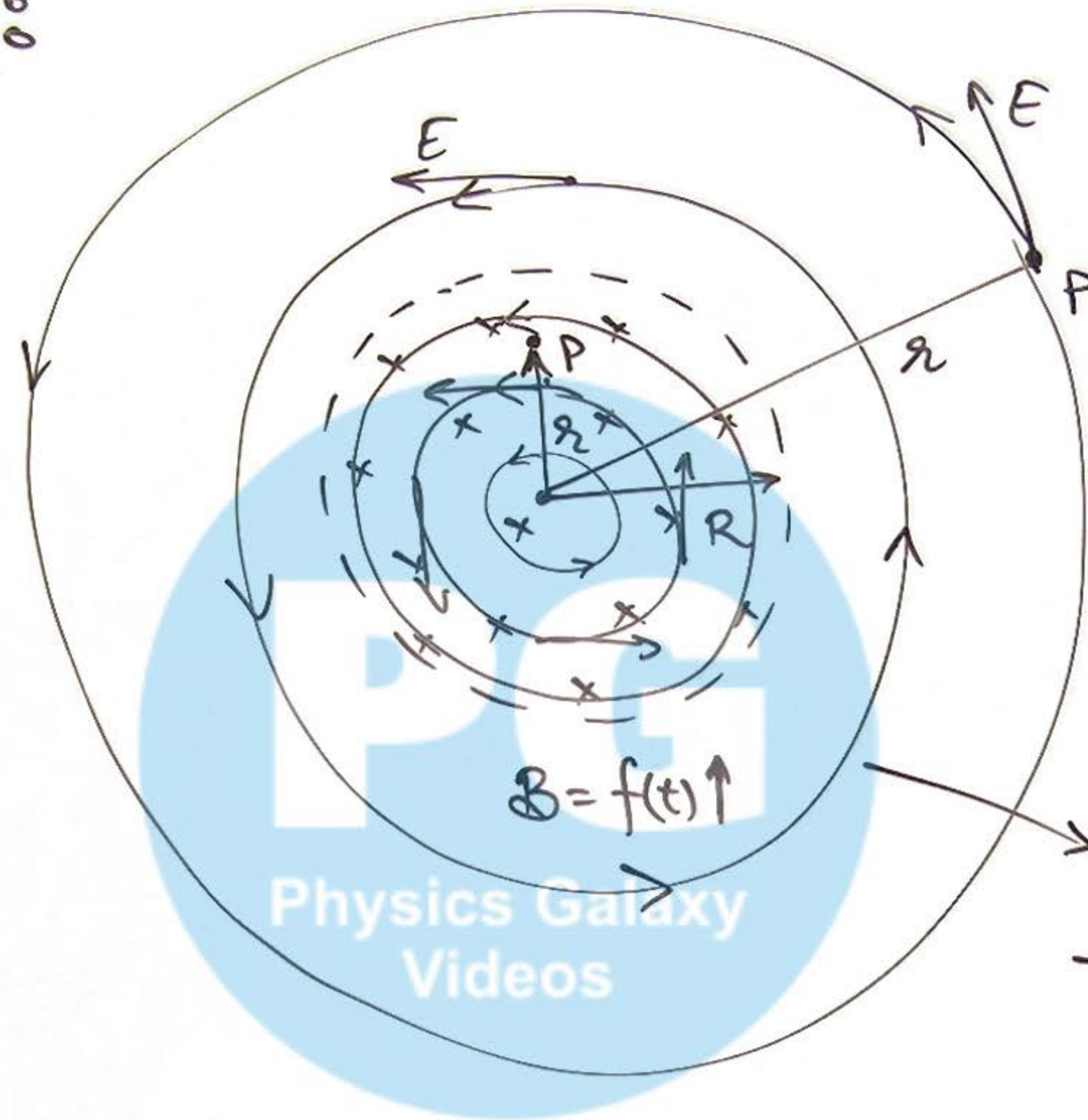


$$e_{\text{loop}} = \int de$$

13 Induced EF in a cylindrical
Region TVMF:

for $r < R$

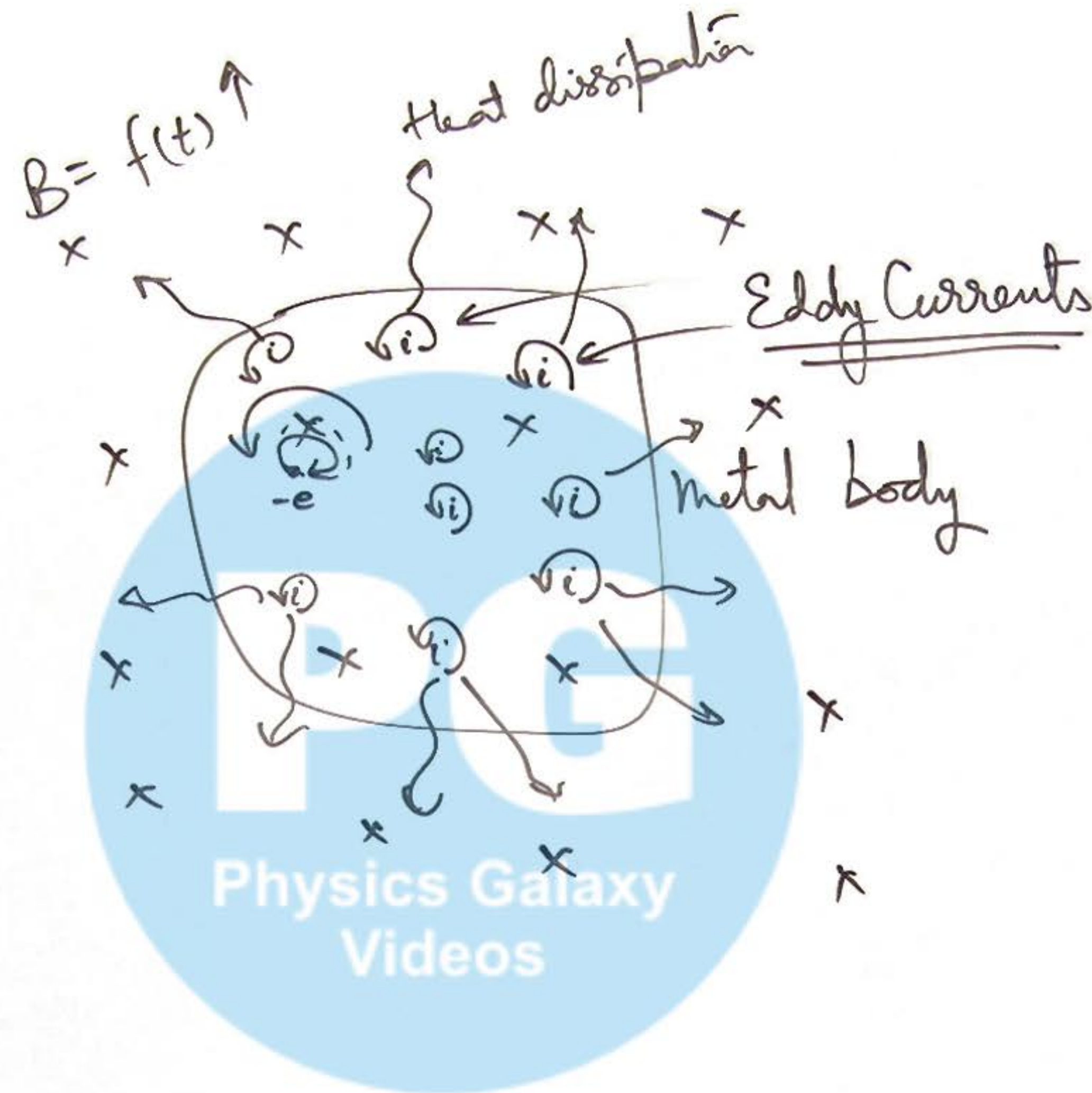
$$\left[\begin{aligned} \mathcal{E}_{in} &= \frac{1}{2} r \frac{dB}{dt} \\ \mathcal{E}_{out} &= \frac{R^2}{2r} \frac{dB}{dt} \end{aligned} \right]$$



Due to closed loop ELF
this EF is non-Conservative.

14 Eddy Currents in TVMF:

Miniature localized
currents in a
Metal Body.



15 Charge flown in a coil due to induced EMF :



emf induced $\mathcal{E} = \left| \frac{d\Phi}{dt} \right|$
 induced current $i = \frac{\mathcal{E}}{R} = \frac{1}{R} \left| \frac{d\Phi}{dt} \right|$

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$$dq = \frac{|\mathcal{E}|}{R} dt$$

$$\Rightarrow \Delta q = \frac{|\Delta \Phi|}{R} \quad \text{v. imp}$$

$$\frac{dq}{dt} = \frac{1}{R} \left| \frac{d\Phi}{dt} \right|$$

(16) Self Induction :

Prop by which a
Conductor/Coil opposes
change in current in
itself.

For any coil carrying current I ,

$$MF \quad B \propto I$$

$$\Rightarrow \text{Magnetic flux } \Phi \propto B$$

$$\Rightarrow \underline{\Phi \propto I}$$

Self linked flux.
with the
Conductor/Coil.

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$$\boxed{\Phi = L I}$$

↓
"Inductance"

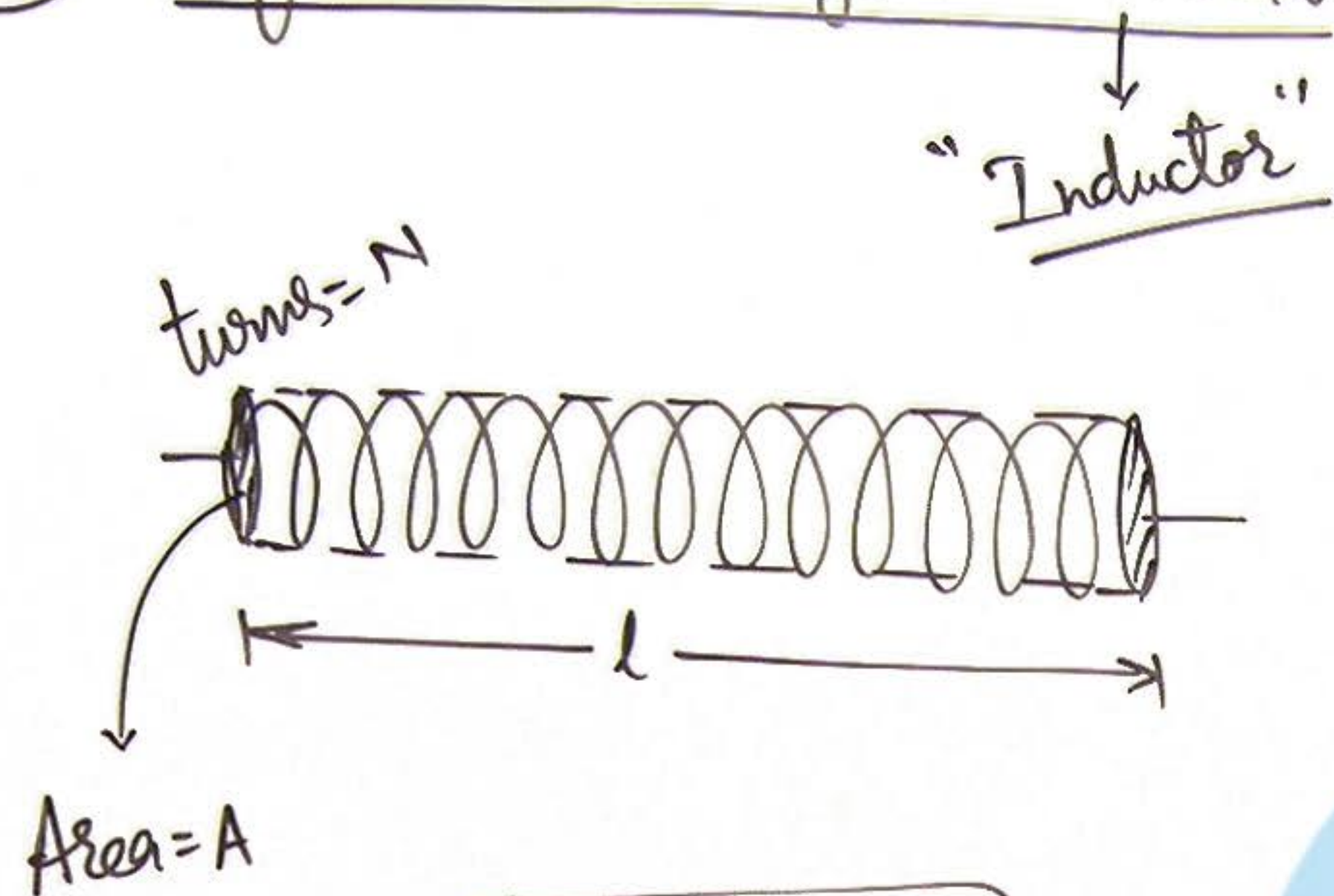
If I changes then
due to own current,
an EMF is induced in
Conductor/Coil -

$$e_s = \left| \frac{d\Phi}{dt} \right| = L \left| \frac{dI}{dt} \right|$$

$$e_s = -L \frac{dI}{dt}$$

↓
Opposition to change in I .

(17) Self Inductance of a Solenoid:



$$L = \frac{\mu_0 N^2 A}{l}$$

Unit $\rightarrow$ "Henry"

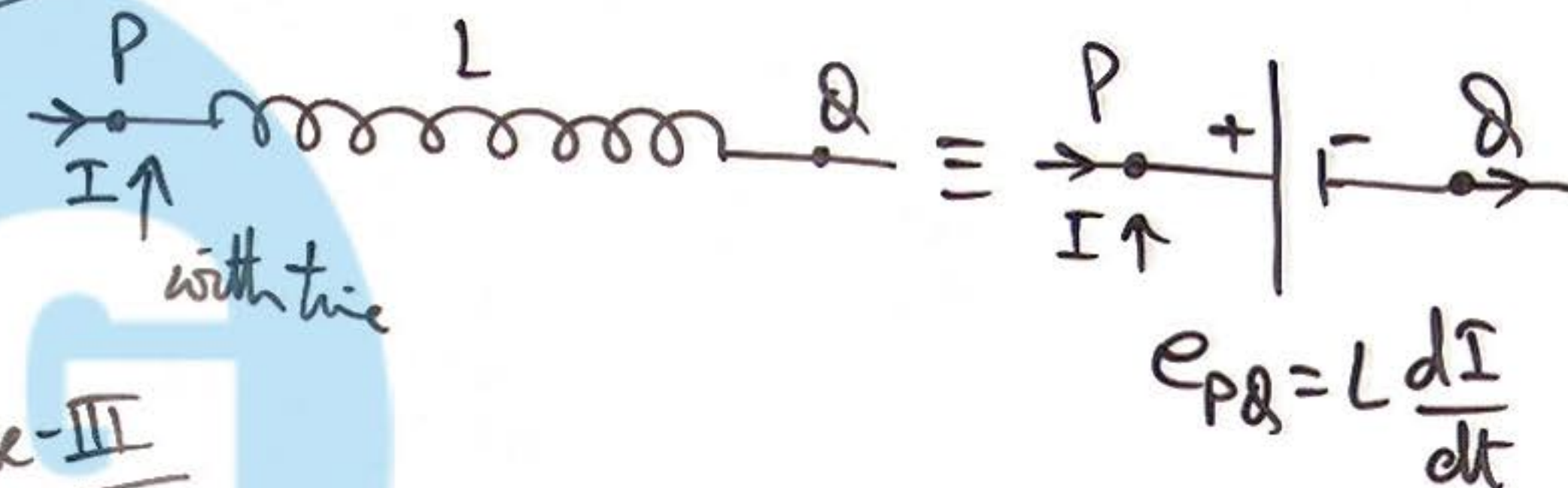
$\times \mu_r$
if core is present

(18) Ckt Behaviour of an Inductor:

Case-I

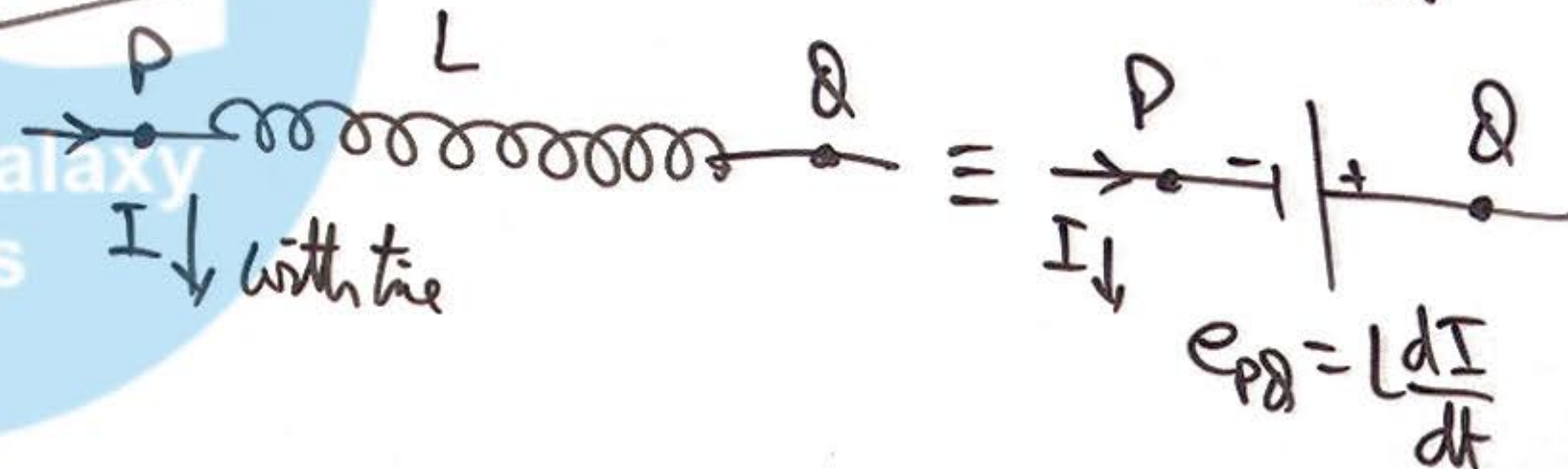


Case-II



$$\mathcal{E}_{PQ} = L \frac{dI}{dt}$$

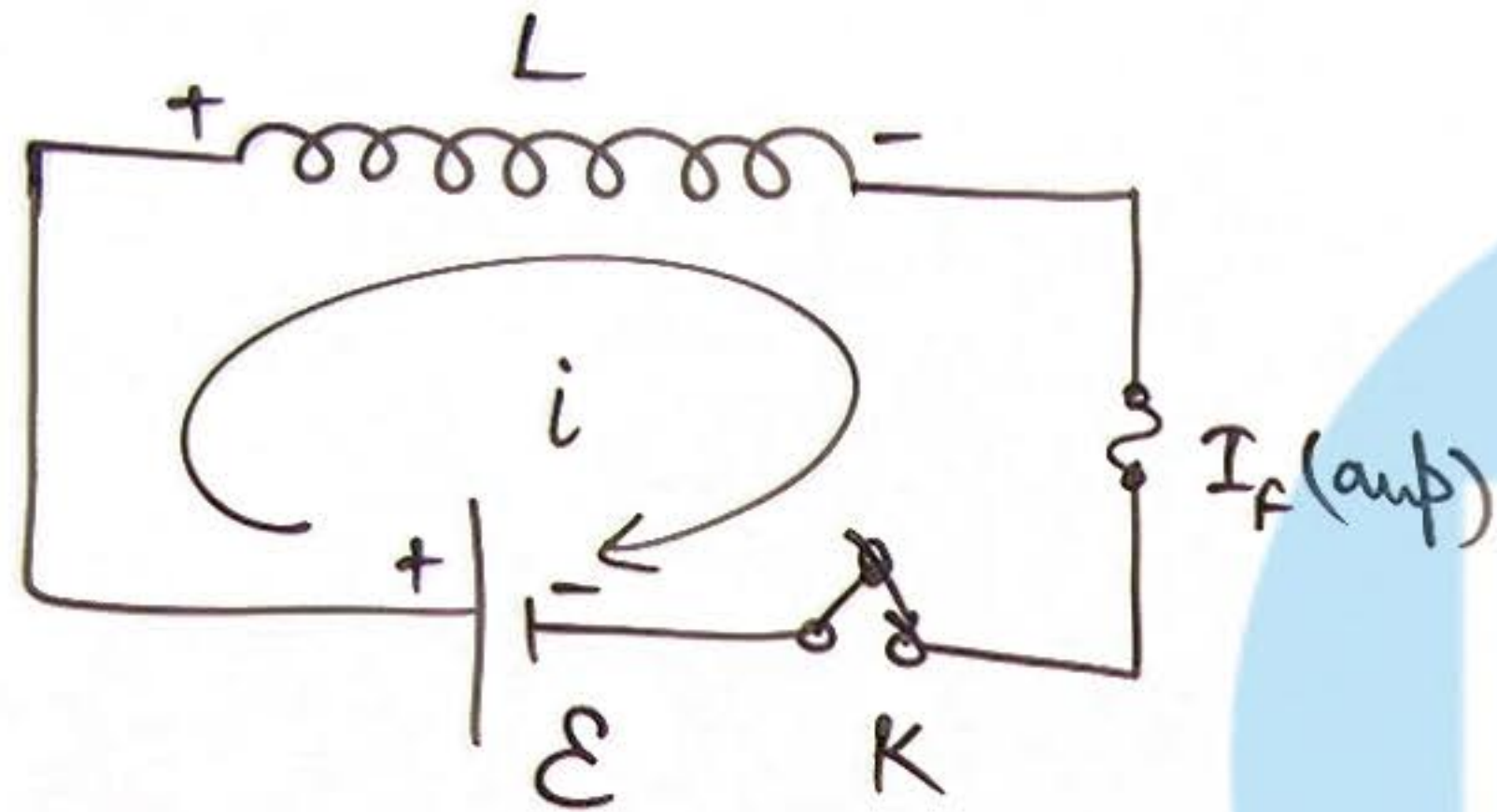
Case-III



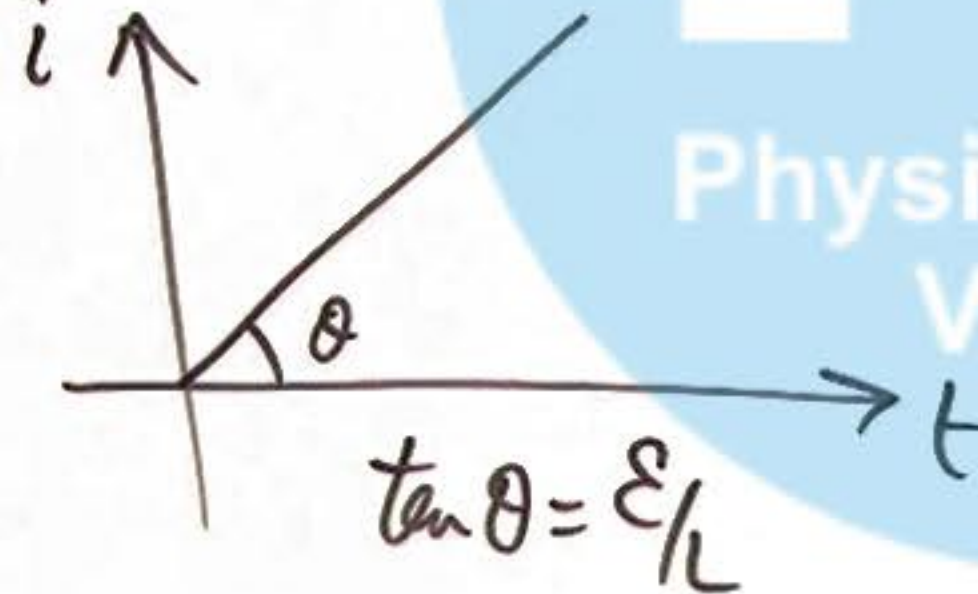
$$\mathcal{E}_{PQ} = L \frac{dI}{dt}$$

(19) Growth of Current in an Inductor:

Case I

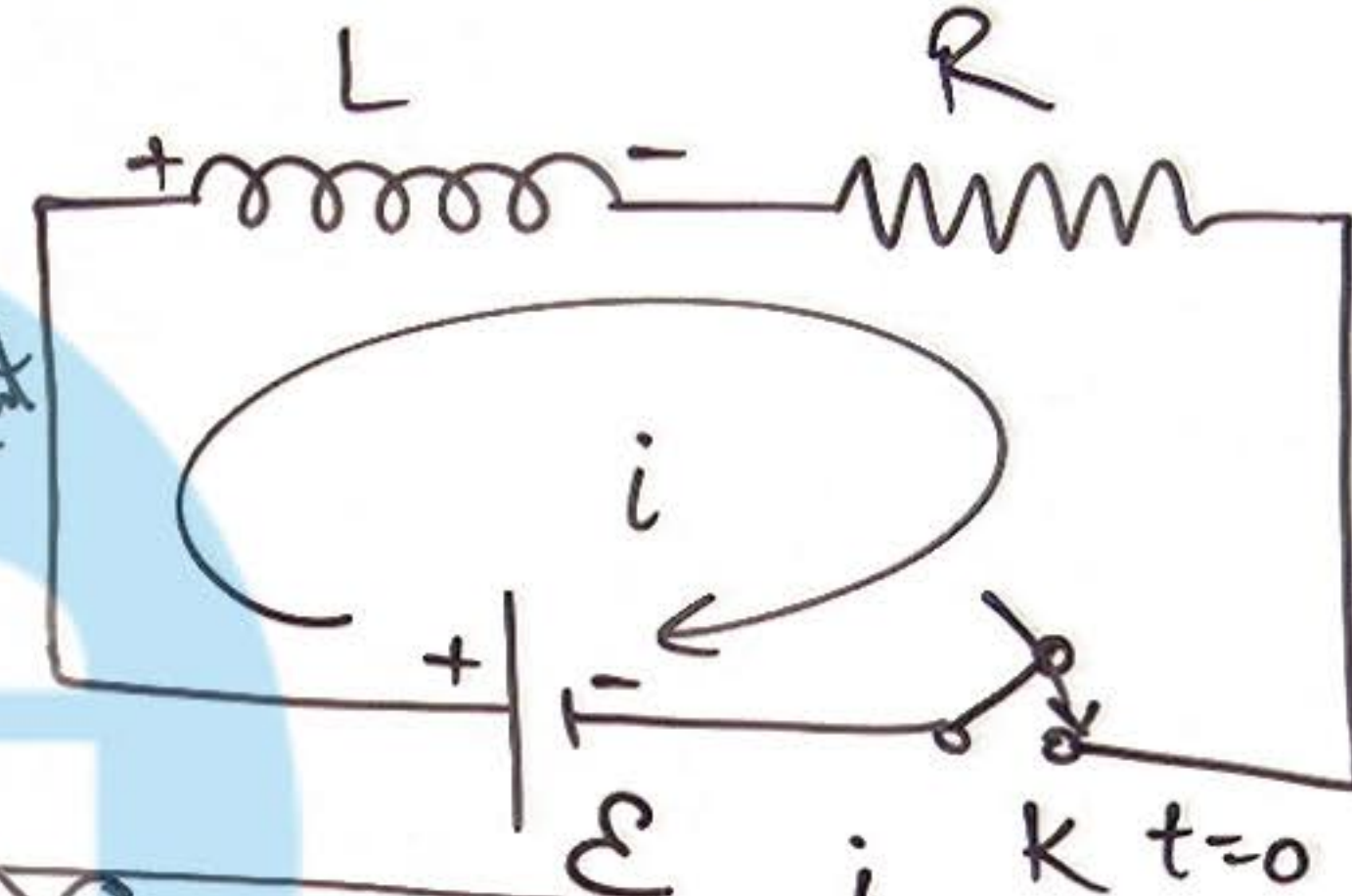


$$i = \frac{\mathcal{E}t}{L}$$



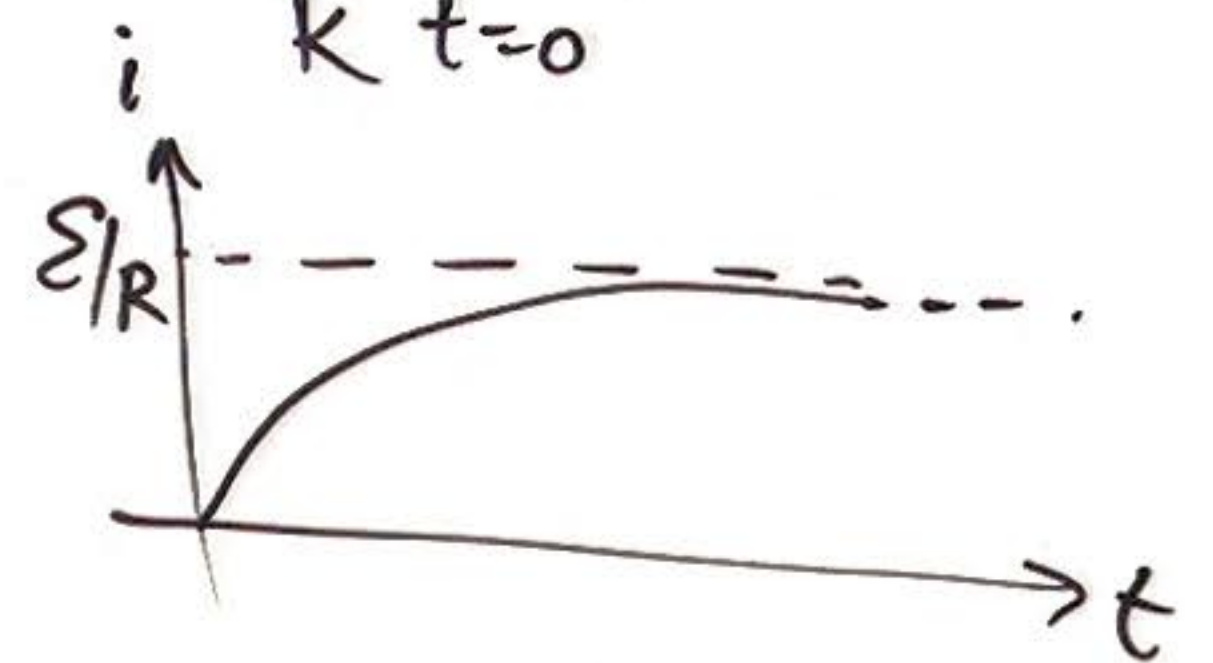
Case II

time constant of RL ckt
 $\tau = \frac{L}{R}$



Steady State current
 $i_0 = \mathcal{E}/R$

$$i = \left(\frac{\mathcal{E}}{R} \right) (1 - e^{-Rt/L})$$

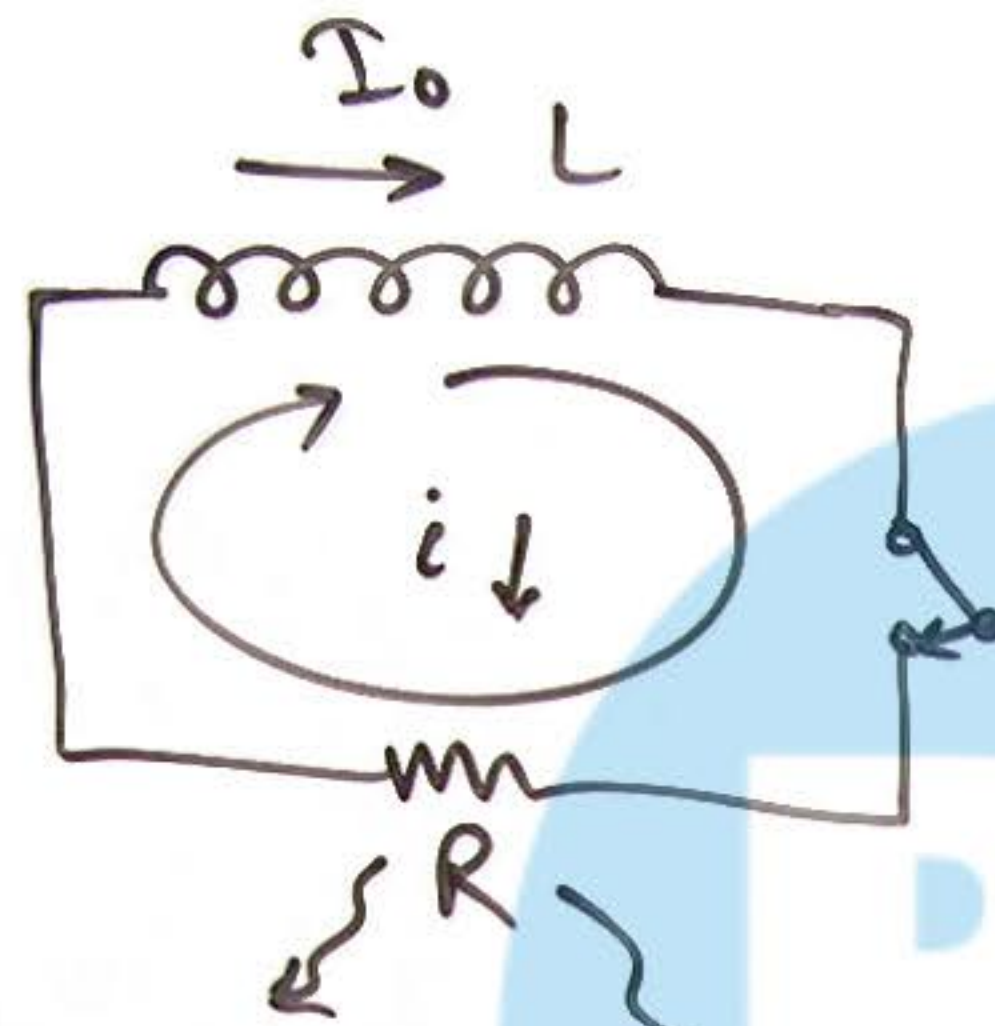


②① Decay of Current in Inductor:

②② Magnetic field
Energy Density:

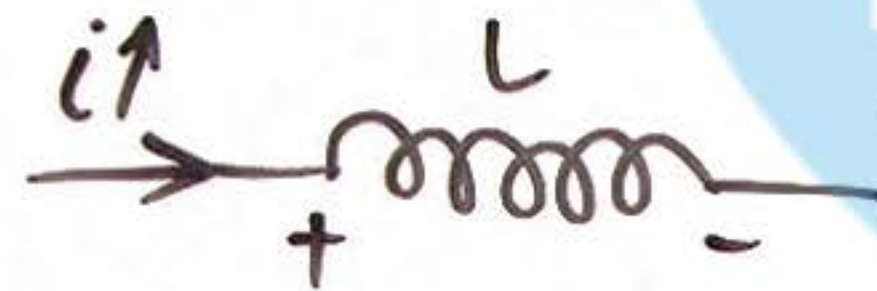
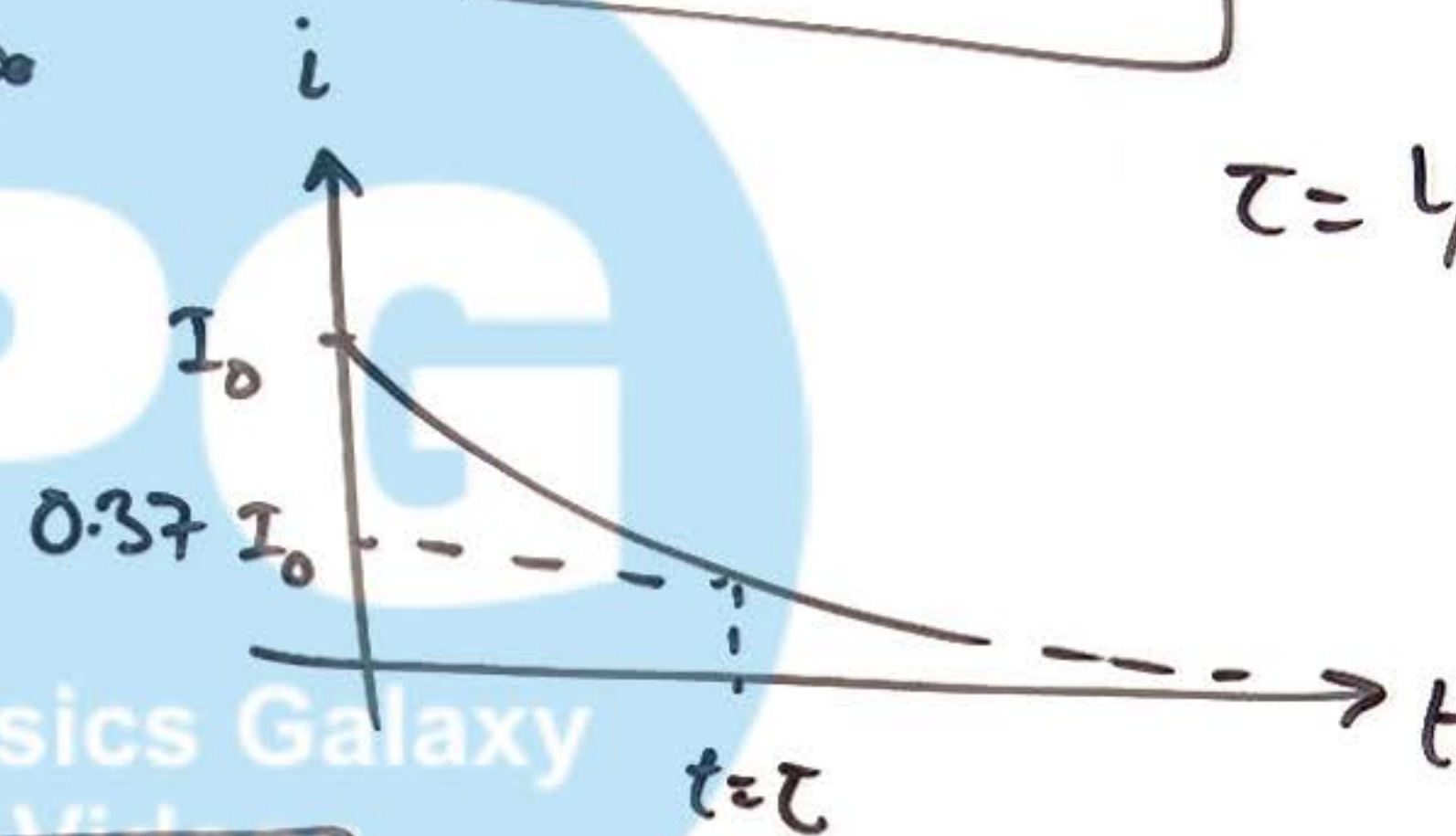
$$U_B = \frac{B^2}{2\mu_0} \text{ J/m}^3$$

②③ Magnetic Energy Stored
in an Inductor:



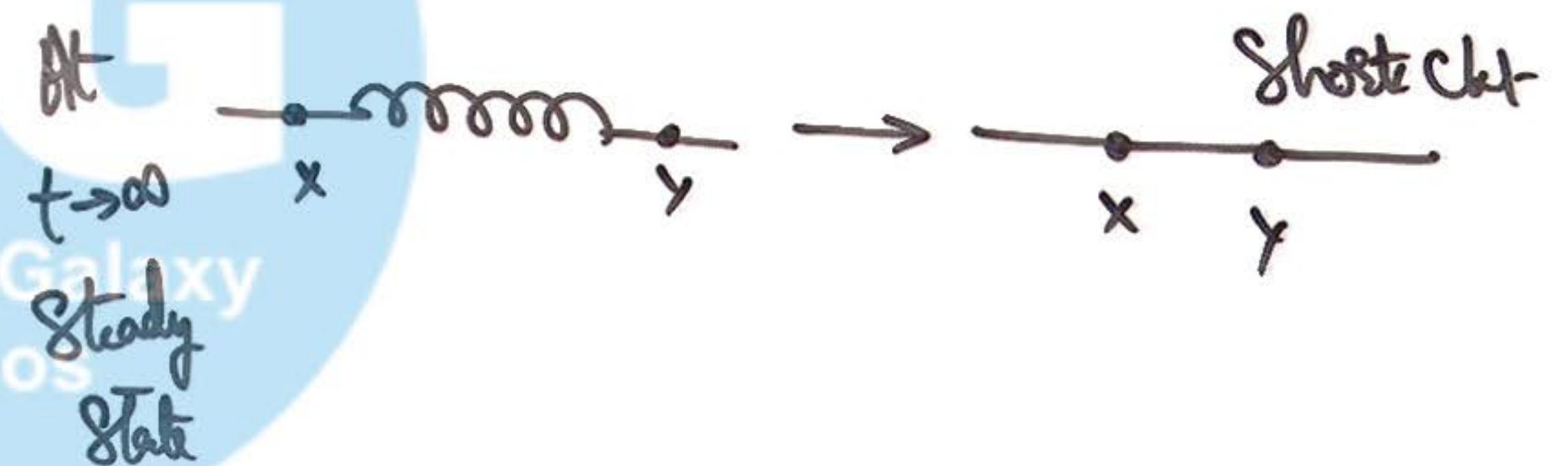
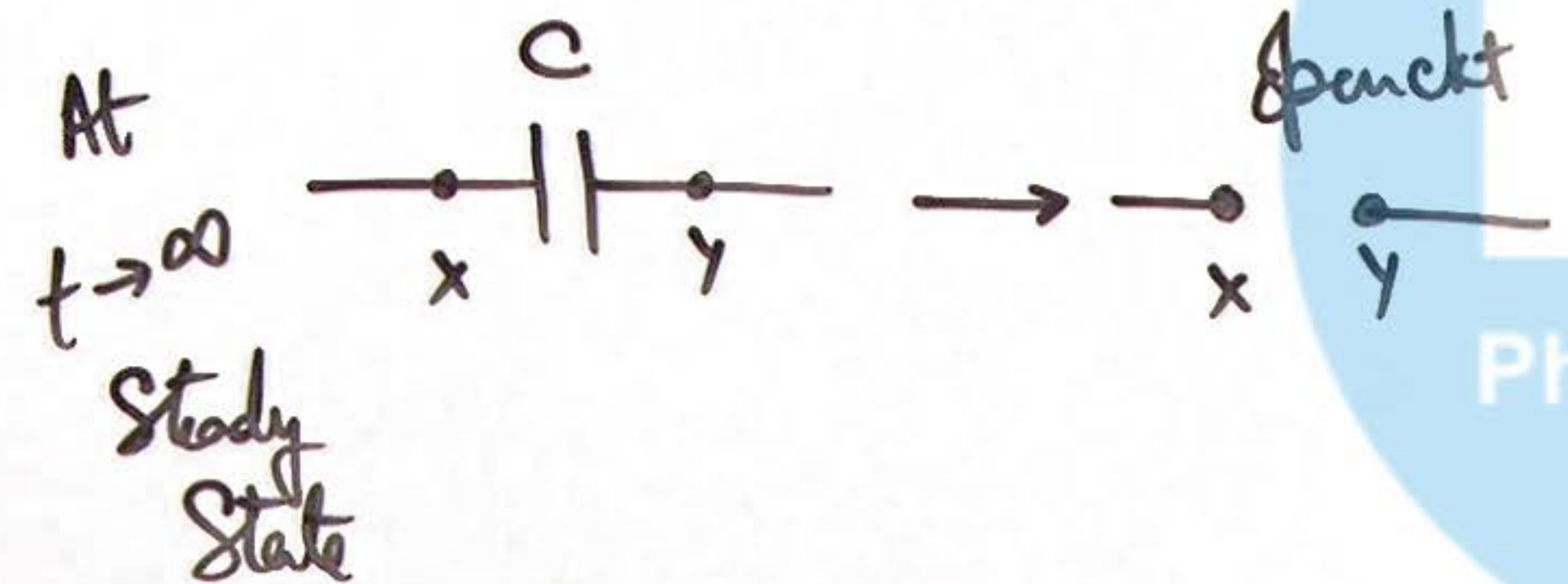
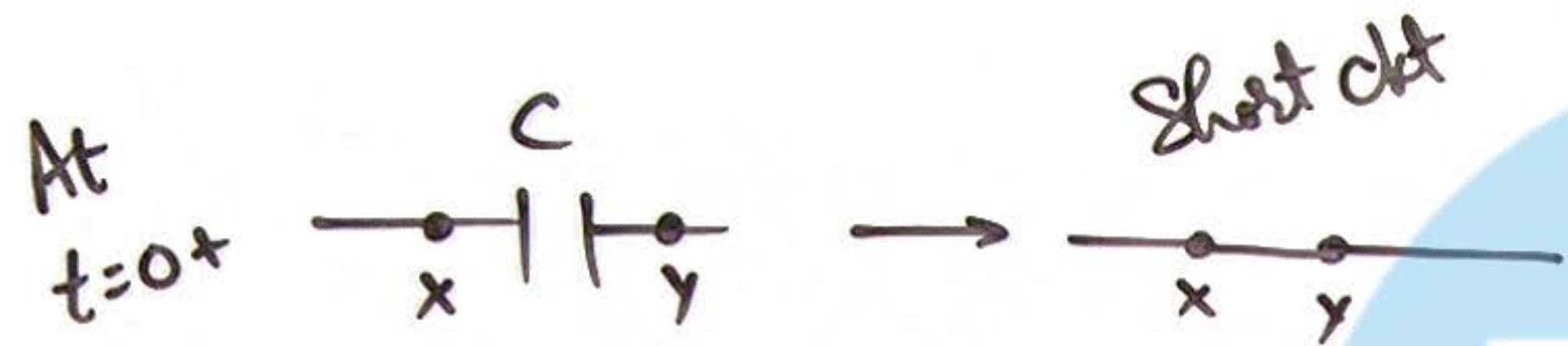
$$i = I_0 e^{-Rt/L}$$

$$\tau = L/R$$



$$U = \frac{1}{2} Li^2$$

(23) Behaviour of Inductor in Switching Circuits:



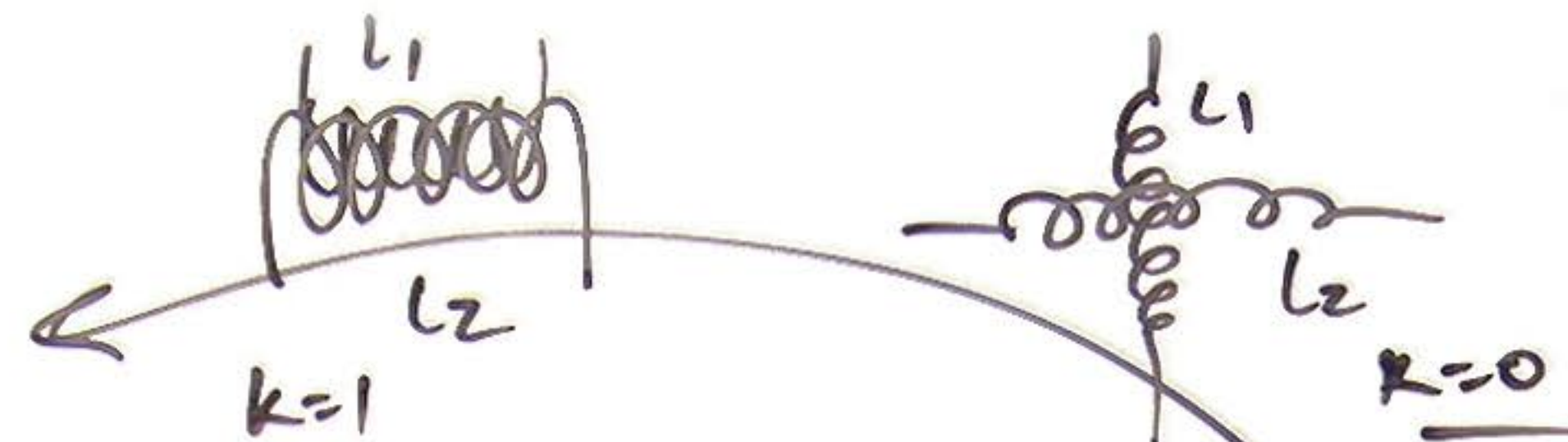
(24) Mutual Induction:

defined for a pair of coil.

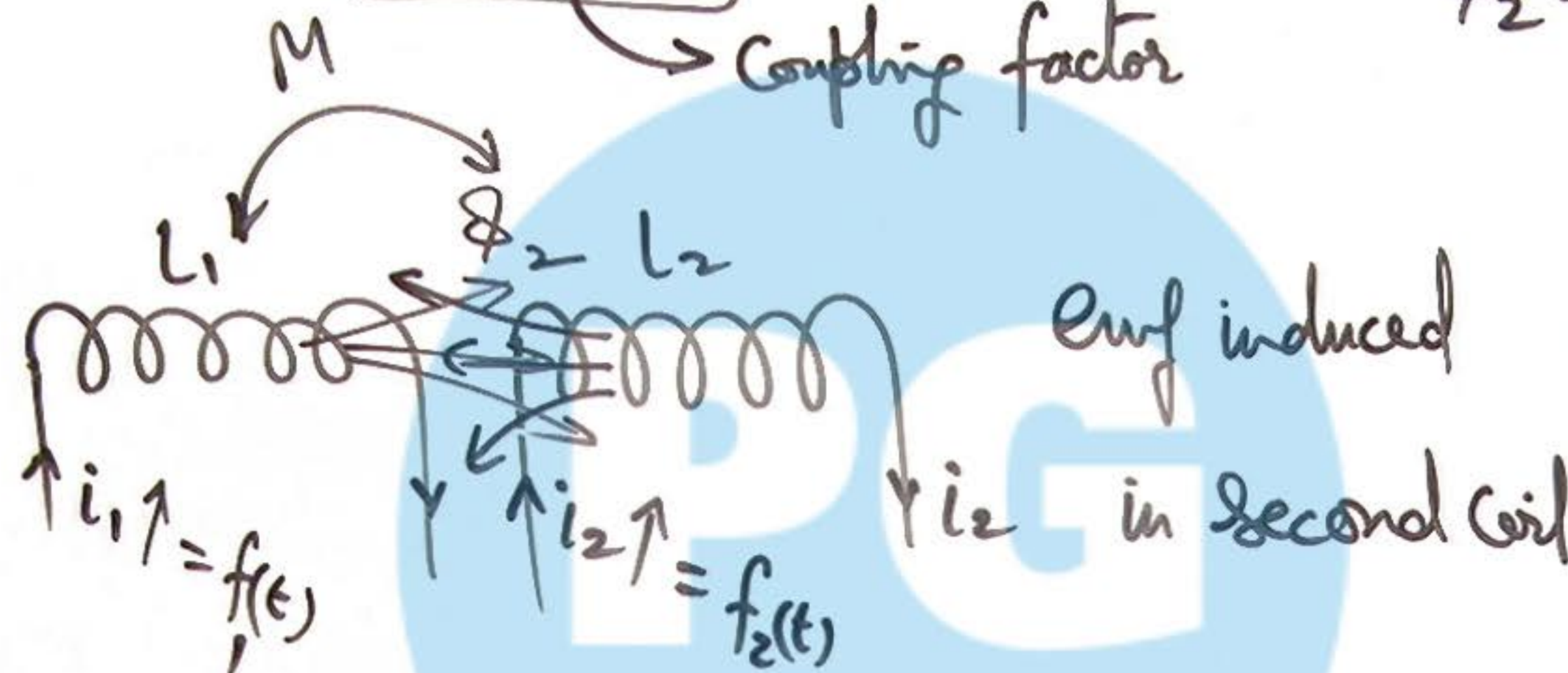
$$M = k \sqrt{L_1 L_2}$$

$$0 \leq k \leq 1$$

Coupling factor



here $\phi_2 \propto i_1$
 $\phi_2 = M i_1$ → Coefficient of Mutual Induction



$$e_2 = \frac{d\phi_2}{dt} = M \frac{di_1}{dt}$$

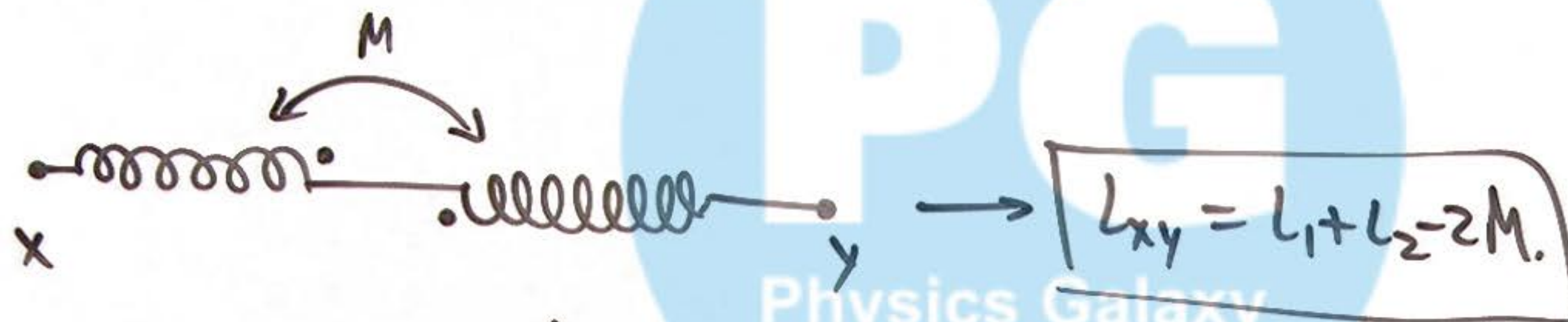
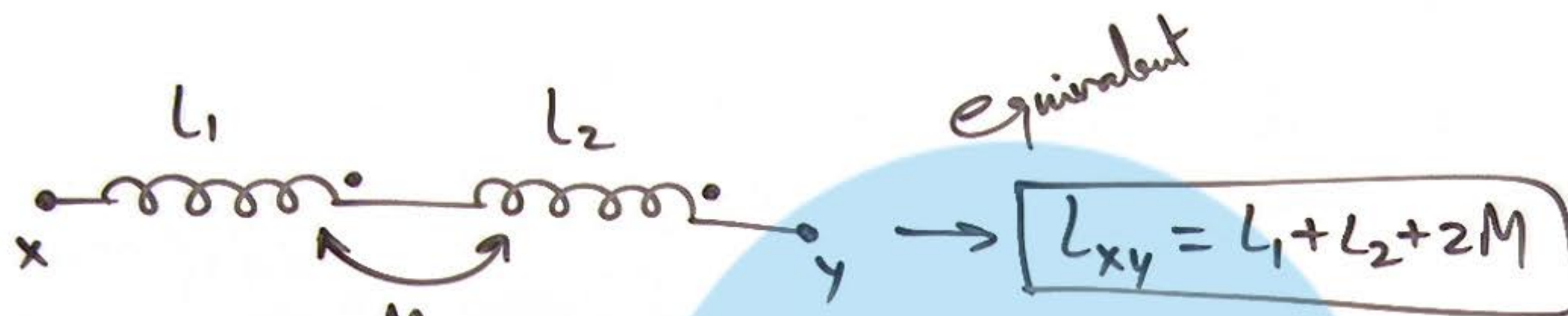
Total induced emf in above coils -

$$e_1 = L_1 \frac{di_1}{dt} \pm M \frac{di_2}{dt} \quad \& \quad e_2 = L_2 \frac{di_2}{dt} \pm M \frac{di_1}{dt}$$

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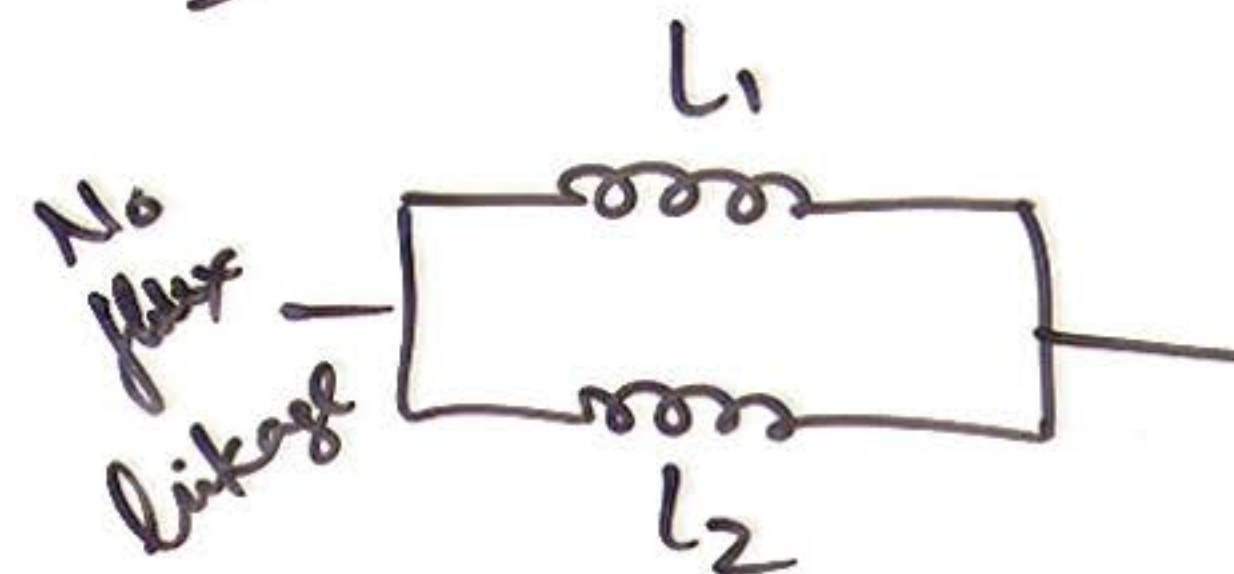
(25) Inductors in Series & Parallel:

Case-I: Series.



for isolated coils in series ($M=0$) $\boxed{L_{xy} = L_1 + L_2}$

Case-II: Parallel.



$$\boxed{\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2}}$$

(26) LC-Oscillations:

Charge & current relation in LC oscillations

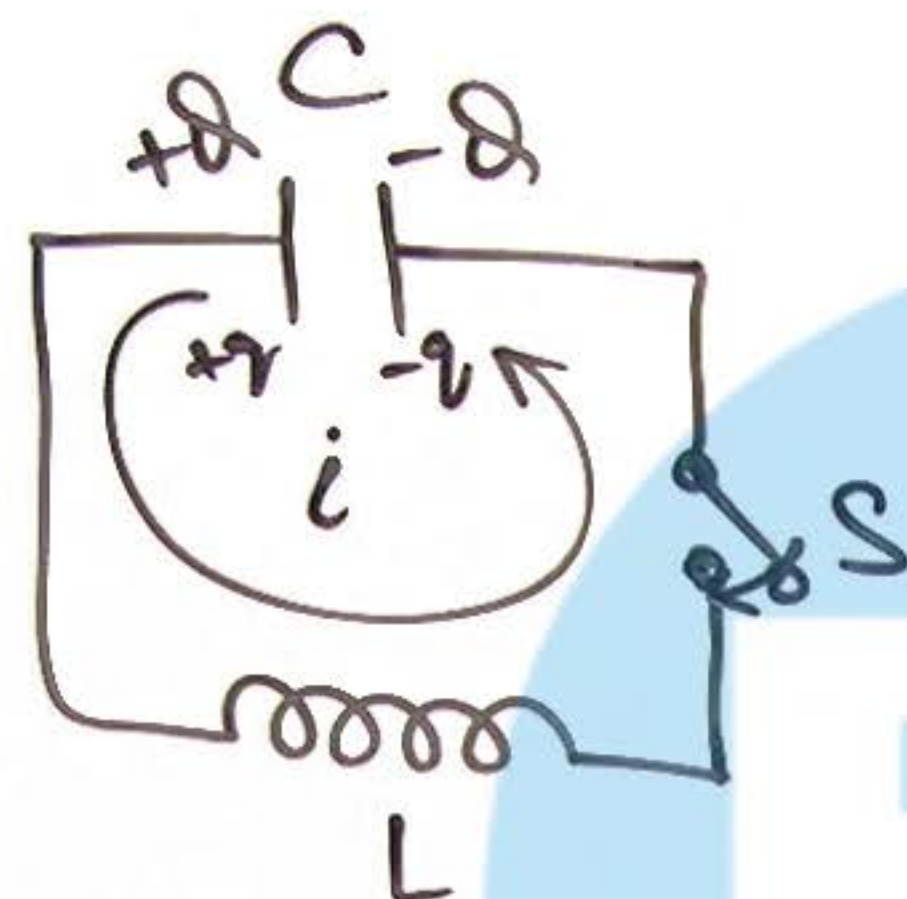
$$q = Q \sin(\omega t + \alpha)$$

$$i = I_0 \cos(\omega t + \alpha)$$

$$i = \frac{1}{\sqrt{LC}} \sqrt{Q^2 - q^2}$$

Ang. freq.

$$\left\{ \begin{array}{l} \omega = \frac{1}{\sqrt{LC}} \\ f = \frac{1}{2\pi\sqrt{LC}} \\ T = 2\pi\sqrt{LC} \end{array} \right.$$



Max Current I_0 is given as

$$\frac{Q^2}{2C} = \frac{1}{2} L I_0^2 \Rightarrow I_0 = Q \sqrt{\frac{1}{LC}}$$

Total energy of LC system

$$E = \frac{q^2}{2C} + \frac{1}{2} L i^2 = \text{const}$$

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